Estimation and Inference for Causal Explainability

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Abstract

Understanding how much each variable contributes to an outcome is a central question across disciplines. A causal view of explainability is favorable for its ability in uncovering underlying mechanisms and generalizing to new contexts. Based on a family of causal explainability quantities, we develop methods for their estimation and inference. In particular, we construct a one-step correction estimator using semi-parametric efficiency theory, which explicitly leverages the independence structure of variables to reduce the asymptotic variance. For a null hypothesis on the boundary, i.e., zero explainability, we show its equivalence to Fisher's sharp null, which motivates a randomization-based inference procedure. Finally, we illustrate the empirical efficacy of our approach through simulations as well as an immigration experiment dataset, where we investigate how features and their interactions shape public opinion toward admitting immigrants.

Keywords: Explainability, Causality, ANOVA, Semi-parametric efficiency, Randomization inference

1. INTRODUCTION

"The truth is rarely pure and never simple." — Oscar Wilde. Outcomes typically rely on multiple factors rather than a single cause: customer ratings are influenced by various attributes of an product; biological traits are influenced by multiple genes (Fisher, 1918; Visscher et al., 2017; Watanabe et al., 2019; Sivakumaran et al., 2011). In the presence of multiple factors, it is practically relevant to quantify their explainability, that is how much a set of factors and their interactions account for the variability in the outcome.

Compared with association-based measures, causality-based explainability can illuminate mechanisms and transfer across contexts reliably, and is sometimes regarded as the foundation of scientific explanation (Pearl, 2009; VanderWeele, 2015). This motivates us to adopt the Causal ANOVA (Gao and Zhao, 2025), a set of quantities generalizing the functional ANOVA (Hooker, 2007) and Sobol indices (Sobol', 2001) to the counterfactual perspective, to measure the explainability. For the statistical analysis of causal ANOVA quantities, prior work has primarily focused on settings in which an oracle[1], typically a black-box machine learning model, is available to evaluate the outcome at arbitrarily chosen input values, with the resulting explainabilities used to assess feature importance. However, in many domains including social science and genetics, researchers rely on collected data samples and have no access to such oracles.

1. For "oracle", we refer to an outcome-generating mechanism: given a user-chosen input vector (e.g., treatment values), one can evaluate the corresponding potential outcome as a function value. This is the standard setup in global sensitivity analysis for black-box simulators, such as Sobol/functional-ANOVA decompositions (Sobol', 2001).

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Semi-parametric methods target the estimation and inference for finite-dimensional parameters while allowing nuisance components to be non-parametric, and operate on pre-collected datasets requiring no oracle. This aligns with our scope: a scalar target (explainability summarized by a number), minimal modeling assumptions, and no oracle access. Therefore, we develop semi-parametric methods for the estimation and inference of Causal ANOVA quantities. Particularly, we derive the efficient influence function and construct the associated one-step correction estimator, and explicitly show how the independence structure in Causal ANOVA can be leveraged to gain efficiency. Degenerate nulls (the parameter of interest lies on the boundary of its space) are a known challenge in semi-parametric statistics (Verdinelli and Wasserman, 2024), and we design a valid randomization-based procedure for such nulls in contrast to the common practice based on sample splitting or noise injection.

Contributions. We investigate the estimation and inference for the explainability attributable to individual factors and their interactions through the lens of semi-parametric statistics.

- Our approach can be used to identify influential factors and interactions by testing for non-zero explainability, and further quantify their importance by providing confidence intervals. Applied to an immigration-experiment dataset, multiple factors are shown to exhibit nonzero causal explainability for public opinions toward admitting immigrants, with interaction results compared against CRT and ACIE analyses in Appendix G.

- We derive the efficient influence function that explicitly leverages the structure, i.e., (conditional) independence, in the probability model and construct the associated one-step correction estimator, which attains no larger asymptotic variance compared to the counterpart that ignores the structure. The benefits of exploiting independence structure as well as our derivation extend to other applications with independent components, such as factorial A/B tests.

- We propose a randomization-based procedure to test whether the null is degenerate, which requires neither sample splitting nor noise injection. Further combined with the confidence interval using the asymptotic distribution of one-step correction estimators (valid under non-degenerate nulls), we introduce a sequential procedure that returns valid confidence sets regardless of the null degeneracy.

Organization. In Section 2, we review Causal ANOVA, semi-parametric estimation theory, and randomization test. In Section 3, we exploit the independence structure in the probability model to derive a more efficient influence function and provide the associated one-step estimator. In Section 4, we derive the one-step estimator's asymptotic distribution under non-degenerate nulls, introduce a randomization-based test for degenerate nulls, and combine them into a sequential procedure. In Section 5, Section 6, we demonstrate our procedure in simulations and a real world dataset. Proofs, additional simulations are provided in the appendix.

Notation. Let ξ denote the causal estimand of interest. Let Y be the outcome and $\boldsymbol{W} = (W_1, \ldots, W_K)$ be the vector of K factors; write $[K] = \{1, \ldots, K\}$ and $\boldsymbol{W}_{\mathcal{S}}$ for the subvector indexed by $\mathcal{S} \subseteq [K]$. For $\mathcal{S} \subseteq [K]$, write $-\mathcal{S} = [K] \setminus \mathcal{S}$ for its complement. Let $\mathbb{P}$ denote the true joint distribution of $(Y, \boldsymbol{W})$, and let $\mu(\boldsymbol{w}) = \mathbb{E}[Y \mid \boldsymbol{W} = \boldsymbol{w}]$ and $\upsilon(\boldsymbol{w}) = \mathbb{E}[Y^2 \mid \boldsymbol{W} = \boldsymbol{w}]$ denote the conditional mean and second moment. Write $\mathbb{P}_{\boldsymbol{W}}$ for the marginal distribution of $\boldsymbol{W}$, $\mathbb{P}_{W_k}$ for the marginal of W_k, and $\mathbb{P}_{Y|\boldsymbol{W}}$ for the conditional distribution of Y given $\boldsymbol{W}$. Their

estimated counterparts are denoted by $\widehat{\mathbb{P}}$, $\widehat{\mu}$ and $\widehat{v}$, with $\widehat{\mathbb{P}}_{\boldsymbol{W}}$, $\widehat{\mathbb{P}}_{W_k}$, and $\widehat{\mathbb{P}}_{Y|\boldsymbol{W}}$ defined analogously. When densities are used, write p_{W_k} and $\widehat{p}_{W_k}$ for the true and fitted marginal densities with respect to a common dominating measure. For a probability model $\mathcal{P}$ and $\mathbb{P} \in \mathcal{P}$, let $\dot{\mathcal{P}}_{\mathbb{P}}$ denote the tangent space at $\mathbb{P}$ (subscript $\mathbb{P}$ dropped when unambiguous). We write $\varphi^{\xi}_{\text{IF}}$, $\varphi^{\xi}_{\text{EIF}}$, or simply φ^{ξ} for influence functions of quantity ξ (superscript ξ dropped when unambiguous). For a more comprehensive summary of the notations, see Table 3.

2. PROBLEM FORMULATION

2.1. Causal ANOVA

Causal ANOVA is set in the potential outcome framework (Rubin, 1974), with K randomized treatments $W_k \in \mathcal{W}_k$ and n units. Each unit i has potential outcomes $\boldsymbol{Y}_i(\cdot)$, but only $Y_i = Y_i(\boldsymbol{W}_i)$ is observed for the realized assignment $\boldsymbol{W}_i = (W_{1,i}, \ldots, W_{K,i})$. To ensure that Causal ANOVA quantities generalize to future observations, we assume the observed units are drawn i.i.d. from a super-population model. Formally,

$$\boldsymbol{W}_i \sim \mathbb{P}_{\boldsymbol{W}},\ \boldsymbol{Y}_i(\cdot) \sim \mathbb{P}_{Y(\cdot)},\ \boldsymbol{W}_i \perp\!\!\!\perp Y_i(\cdot), \tag{1}$$

independently across units, where $\mathbb{P}_{\boldsymbol{W}}$ is the (unknown) distribution of treatment assignments and $\mathbb{P}_{Y(\cdot)}$ is the unknown distribution of potential outcomes.

Causal ANOVA defines explainability via contrasts among potential outcomes, i.e., comparing factuals and counterfactuals. While Causal ANOVA can accommodate any set of factors $\boldsymbol{W}$ that respect a directed acyclic graph (DAG), identification of the explainabilities under dependent factors generally requires additional assumptions (e.g., comonotonicity of potential outcomes). As a result, in this paper we focus on independent treatments specified below.

Assumption 1 (Independent treatments) *The factors W_k, $k \in [K]$, are mutually independent.*

Let $\mathcal{P}_{\text{ind}}$ denote the model satisfying Assumption 1, a proper subset of the non-parametric model $\mathcal{P}_{\text{np}}$ in (1). Assumption 1 is satisfied in many experiment designs, for example, conjoint analysis randomizes the value of attributes independently. More generally, settings with block-wise independence of treatment groups can be analyzed in a similar manner, where each block of treatments is regarded as a single combined factor. In Appendix A, we show Assumption 1 can be relaxed to the conditional independence: W_k are mutually independent given some covariates $\boldsymbol{X}$.

Let $\boldsymbol{W}'$ have the same law as $\boldsymbol{W}$ and be independent of $(\boldsymbol{W}, Y(\cdot))$. Throughout, assume $Y \in L_2$ and $0 < \mathsf{Var}(Y) < \infty$. Under Assumption 1, Causal ANOVA uses the variance of the difference obtained by replacing $\boldsymbol{W}_{\mathcal{S}}$ with $\boldsymbol{W}'_{\mathcal{S}}$ to define the total explainability of $\boldsymbol{W}_{\mathcal{S}}$, $\mathcal{S} \subseteq [K]$.

Definition 1 (Total explainability) *For $\mathcal{S} \subseteq [K]$,*

$$\xi(\vee_{k\in\mathcal{S}} W_k) := \frac{\mathsf{Var}\left(Y(\boldsymbol{W}) - Y(\boldsymbol{W}'_{\mathcal{S}}, \boldsymbol{W}_{-\mathcal{S}})\right)}{2\mathsf{Var}\left(Y(\boldsymbol{W})\right)}. \tag{2}$$

The explainability of interaction is defined as the variance of a difference-in-difference contrast.

Definition 2 (Interaction explainability) *For $k \neq k' \in [K]$,*

$$\xi(W_k \wedge W_{k'}) := \frac{\mathsf{Var}\left(I(W_k, W_{k'})\right)}{4\,\mathsf{Var}\left(Y(\boldsymbol{W})\right)}, \tag{3}$$

where $I(W_k', W_{k'}') = Y(W_k', W_{k'}') - Y(W_k', W_{k'}) - Y(W_k, W_{k'}') + Y(W_k, W_{k'})$. *Here the remaining treatments are held at* $\boldsymbol{W}_{-\{k,k'\}}$.

Definition 2 defines the first-order interaction, and can be generalized to higher-order interactions. However, we omit them due to the reduced usefulness and increased complexity.

When Y is not fully determined by $\boldsymbol{W}$, the Causal ANOVA quantities are generally not identifiable. The non-parametric structural equation model with additive, independent errors, a.k.a. NPSEM-IE (Pearl, 2009), offers a useful balance between identifiability and generality.

Assumption 2 (NPSEM-IE) *The potential outcomes satisfy* $Y(\boldsymbol{w}) = f_Y(\boldsymbol{w}) + E_Y$, *with the same unit-level error* E_Y *across treatment levels and* E_Y *independent of* $\boldsymbol{W}$. *The observed outcome is* $Y = Y(\boldsymbol{W})$.

NPSEM-IE is a commonly used model for causal discovery and inference (Hoyer et al., 2008; Peters et al., 2014). We discuss its extensions in Appendix A.2.

Proposition 1 *Under Assumption 1 and Assumption 2, for any* $\mathcal{S} \subseteq [K]$, *Definition 1 is identifiable and admits the form,*

$$\xi(\vee_{k\in\mathcal{S}} W_k) = \frac{\mathbb{E}[\mathbb{E}^2[Y \mid \boldsymbol{W}]]}{\mathsf{Var}(Y)} - \frac{\mathbb{E}\left[\mathbb{E}^2[Y \mid \boldsymbol{W}_{-\mathcal{S}}]\right]}{\mathsf{Var}(Y)}. \tag{4}$$

By the inclusion-exclusion principle for causal explainabilities, that is $\xi(W_k \wedge W_{k'}) = \xi(W_k) + \xi(W_{k'}) - \xi(W_k \vee W_{k'})$, we obtain an analogue of Proposition 1 for the interaction explainabilities.

Corollary 1 *Under the assumption of Proposition 1, for* $k \neq k' \in [K]$, *Definition 2 is identifiable and admits the form,*

$$\begin{aligned}\xi(W_k \wedge W_{k'}) = &\frac{\mathbb{E}[\mathbb{E}^2[Y \mid \boldsymbol{W}]]}{\mathsf{Var}(Y)} - \frac{\mathbb{E}\left[\mathbb{E}^2[Y \mid \boldsymbol{W}_{-k}]\right]}{\mathsf{Var}(Y)} \\ &- \frac{\mathbb{E}\left[\mathbb{E}^2[Y \mid \boldsymbol{W}_{-k'}]\right]}{\mathsf{Var}(Y)} + \frac{\mathbb{E}\left[\mathbb{E}^2[Y \mid \boldsymbol{W}_{-\{k,k'\}}]\right]}{\mathsf{Var}(Y)}.\end{aligned} \tag{5}$$

2.2. Influence function and one-step correction

Influence functions are critical in semi-parametric inference for constructing approximately unbiased and efficient estimators robust to nuisance estimation (Bickel et al., 1993). Let the quantity of interest ξ be some functional of $\mathbb{P}$, e.g., the causal explainability for a set of factors. Let $\mathcal{P}$ be a class of probabilities for the law $\mathbb{P}$ of $(Y_i, \boldsymbol{W}_i)$, and define its tangent space $\dot{\mathcal{P}}$ as the closure of the linear span of its scores. Suppose the functional ξ admits the von Mises expansion with remainder R_2:

$$\xi(\mathbb{P}_\varepsilon) - \xi(\mathbb{P}) = \int \varphi(y, \boldsymbol{w}; \mathbb{P}_\varepsilon)\, d(\mathbb{P}_\varepsilon - \mathbb{P})(y, \boldsymbol{w}) + R_2(\mathbb{P}_\varepsilon, \mathbb{P}), \tag{6}$$

for a perturbed distribution $\mathbb{P}_\varepsilon$ of $\mathbb{P}$, then $\varphi(y, \boldsymbol{w}; \mathbb{P})$ is a mean-zero, finite-variance function satisfying $\int \varphi(y, \boldsymbol{w}; \mathbb{P}) d\mathbb{P}(y, \boldsymbol{w}) = 0$, typically called an influence function. Influence functions are not unique; however, the projection of any influence function onto the tangent space $\dot{\mathcal{P}}$ is unique, which is called the *efficient influence function*, denoted by $\varphi_{\text{EIF}}(y, \boldsymbol{w}; \mathbb{P})$, and the semi-parametric

efficiency bound of ξ is $\mathsf{Var}(\varphi_{\text{EIF}}(Y, \boldsymbol{W}; \mathbb{P}))$. We emphasize that for the same functional, the efficient influence function depends on the chosen model class $\mathcal{P}$. A smaller class $\mathcal{P}$ yields a smaller tangent space $\dot{\mathcal{P}}$ and hence likely a lower semi-parametric efficiency bound.

Influence functions provide a principled way to correct plug-in estimators and construct less biased, more efficient estimators. Particularly, let $\widehat{\mathbb{P}}$ be an estimator of the true distribution $\mathbb{P}$. A natural plug-in estimator of $\xi(\mathbb{P})$ is $\xi(\widehat{\mathbb{P}})$. Given an influence function φ, the associated one-step corrected estimator is

$$\widehat{\xi} = \xi(\widehat{\mathbb{P}}) + \mathbb{P}_n \widehat{\varphi}, \quad \mathbb{P}_n \widehat{\varphi} := \frac{1}{n} \sum_{i=1}^{n} \varphi(Y_i, \boldsymbol{W}_i; \widehat{\mathbb{P}}). \tag{7}$$

If $\xi(\widehat{\mathbb{P}})$ is regular and efficient, then the one-step corrected estimator $\widehat{\xi}$ is also efficient, in the sense of attaining the semi-parametric efficiency bound.

We also review the literature on degenerate nulls (the parameter of interest lies on the boundary of the parameter space). For degenerate nulls, the influence functions of the estimator vanish. As a result, the estimator typically converges to a degenerate distribution, which can not be used to construct valid confidence intervals. Currently there are three major approaches to addressing the above problem (Dai et al., 2022; Williamson et al., 2023; Verdinelli and Wasserman, 2024). As noted by Verdinelli and Wasserman (2024), all these approaches rely on $O(n^{-1/2})$-level expansions to preserve validity under the null, at the cost of efficiency. In Hudson (2023), an estimator is proposed, which converges at rate n^{-1} under the null and at rate $n^{-1/2}$ away from the null. However, the method does not yield valid confidence intervals in the null hypothesis's neighborhood.

2.3. Fisher's randomization test

Fisher's randomization test originally evaluates a test statistic over the randomization distribution induced by the assignment mechanism under some sharp null (Fisher, 1918). Explicitly, given a test statistic T, we first compute $T(Y, \boldsymbol{W})$ under the observed assignment $\boldsymbol{W}$. Under the sharp null, we impute the potential outcomes under the partially sharp null, draw assignments $\boldsymbol{W}^{(b)}$ independently from the known conditional assignment mechanism, using the same conditioning as for the observed assignment, evaluate $T(Y, \boldsymbol{W}^{(b)})$ to form the randomization distribution, and compute the p-value

$$P := \frac{1 + \sum_{b=1}^{B} \mathbf{1}\{\, T(Y, \boldsymbol{W}^{(b)}) \geq T(Y, \boldsymbol{W}) \,\}}{1 + B}. \tag{8}$$

When the observed and resampled statistics are jointly exchangeable under the null, any finite B gives a finite-sample valid p-value in the sense that $\mathbb{P}(P \leq \alpha) \leq \alpha$ for all $\alpha \in (0, 1)$ (Phipson and Smyth, 2010). In Causal ANOVA, the degenerate null (explainability being zero) implies Fisher's sharp null (Section 4), motivating the use of randomization-based inference.

3. ESTIMATION

We build on the semi-parametric efficiency theory to derive the influence function for the Causal ANOVA quantities (1) and then incorporate it into the standard cross-fitting procedure (Algorithm 1) (Schick, 1986; Klaassen, 1987) to obtain the one-step correction estimator (Pfanzagl, 1990; Bickel et al., 1993). Among the one-step correction estimators satisfying the asymptotic distribution established in Section 4 below, the one with a smaller asymptotic variance, i.e., higher efficiency, is

preferred. In particular, the influence function that yields an estimator attaining the semi-parametric efficiency bound regarding a model $\mathcal{P}$ ($\mathcal{P}$ denoting a class of candidate joint distributions of $(Y, \boldsymbol{W})$) is optimal and referred to as the efficient influence function (for $\mathcal{P}$).

According to Proposition 1 and Corollary 1, for the total or interaction explainabilities, it suffices to estimate quantities of the form $\mathbb{E}\left[\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right]/\mathsf{Var}\,(Y)$, $\mathcal{S} \subseteq [K]$, which is pathwise differentiable (Williamson et al., 2021).

3.1. Baseline estimator

To begin with, consider the non-parametric model $\mathcal{P}_{\text{np}}$ for $(Y, \boldsymbol{W})$. Under this model, the efficient influence function for $\mathbb{E}[\mathbb{E}(Y \mid \boldsymbol{W}_{-\mathcal{S}})^2]/\mathsf{Var}(Y)$ takes the form (derivation in Appendix D.2),

$$\varphi_{\text{IF}}(Y, \boldsymbol{W}; \mathbb{P}) := \frac{2Y\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}] - \mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]^2}{\mathsf{Var}\,(Y)} - \mathbb{E}\left[\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right]\left(\frac{Y - \mathbb{E}\,[Y]}{\mathsf{Var}\,(Y)}\right)^2. \tag{9}$$

3.2. Estimator leveraging independence

We leverage the independence among W_k (1) to refine φ_{IF} in (9) to gain efficiency. The key step is to project φ_{IF} onto the tangent space of the restricted model $\dot{\mathcal{P}}_{\text{ind}}$, a subspace of $\dot{\mathcal{P}}_{\text{np}}$, to get the efficient influence function φ_{EIF}. The difference $\varphi_{\text{IF}} - \varphi_{\text{EIF}}$, while may be influential under $\mathcal{P}_{\text{np}}$, has zero influence on the target estimand when W_k are independent, and variation along these directions can inflate the variability. By removing them, we do not lose any information of the target estimand under the independence (1) and obtain no larger asymptotic variance. Remark 1 provides an alternative explanation that demonstrates the projection procedure and the resulting efficiency gain. The next result shows the efficient influence function incorporating the independence (1).

Proposition 2 *Under the assumptions of Proposition 1, fix any $\mathcal{S} \subseteq [K]$. Assume $0 < \mathsf{Var}(Y) < \infty$ and that $2Y\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}] - \mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]^2 - \mathbb{E}[\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]^2]$ and $(Y - \mathbb{E}Y)^2 - \mathsf{Var}(Y)$ belong to $L_2(\mathbb{P})$. The square-integrability conditions hold, in particular, if $\mathbb{E}[|Y|^4] < \infty$. The efficient influence function of $\mathbb{E}\left[\mathbb{E}\,[Y \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right]/\mathsf{Var}\,(Y)$ in the model with unrestricted conditional law $\mathbb{P}_{Y|\boldsymbol{W}}$ and independent, otherwise unrestricted treatment marginals, i.e., $\mathbb{P}_{\boldsymbol{W}} = \prod_{k\in[K]} \mathbb{P}_{W_k}$, takes the form*

$$\begin{aligned}
\varphi_{EIF}(Y, \boldsymbol{W}; \mathbb{P}) =& \frac{2\,(Y - \mathbb{E}[Y \mid \boldsymbol{W}]) \cdot \mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]}{\mathsf{Var}\,(Y)} \\
&+ \frac{2\sum_{k\in\mathcal{S}} \mathbb{E}\left[\mathbb{E}[Y \mid \boldsymbol{W}] \cdot \mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}] \mid W_k\right]}{\mathsf{Var}\,(Y)} \\
&+ \frac{\sum_{k\in-\mathcal{S}} \mathbb{E}\left[\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]^2 \mid W_k\right]}{\mathsf{Var}\,(Y)} - \frac{(2|\mathcal{S}| + |-\mathcal{S}|) \cdot \mathbb{E}\left[\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right]}{\mathsf{Var}(Y)} \\
&- \frac{\varphi_{EIF}^{\mathsf{Var}(Y)}(Y, \boldsymbol{W}; \mathbb{P}) \cdot \mathbb{E}\left[\mathbb{E}\,[Y \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right]}{\mathsf{Var}\,(Y)^2},
\end{aligned} \tag{10}$$

$$\begin{aligned}
\varphi_{EIF}^{\mathsf{Var}(Y)}(Y, \boldsymbol{W}; \mathbb{P}) =& (Y - \mathbb{E}\,[Y])^2 - \mathsf{Var}\,(Y \mid \boldsymbol{W}) - (\mathbb{E}[Y \mid \boldsymbol{W}] - \mathbb{E}[Y])^2 \\
&+ \sum_{k\in[K]} \mathbb{E}\left[(Y - \mathbb{E}[Y])^2 \mid W_k\right] - K \cdot \mathsf{Var}\,(Y)\,.
\end{aligned} \tag{11}$$

The proof of Proposition 2 relies on explicitly characterizing the tangent space $\dot{\mathcal{P}}_{\text{ind}}$ under (1) and then performing the projection. Details are provided in Appendix D. In Section 5, we demonstrate the efficiency gain of φ_{EIF} over φ_{IF} in simulated datasets.

Remark 1 *According to Proposition 1. in Chapter 3 of Bickel et al. (1993), the efficient influence function can be decomposed as* $\varphi_{EIF} = \varphi - \Pi\{\varphi \mid \dot{\mathcal{P}}_{\text{ind}}^{\perp}\}$*, where* φ *is any influence function and* φ_{EIF} *is orthogonal to* $\Pi\{\varphi \mid \dot{\mathcal{P}}_{\text{ind}}^{\perp}\}$*. The estimator based on* φ_{IF}*,* φ_{EIF} *has asymptotic variance* $\mathsf{Var}(\varphi_{IF})$*,* $\mathsf{Var}(\varphi_{EIF})$*, respectively; for* $\mathbb{P} \in \mathcal{P}_{\text{ind}}$*,* $\mathsf{Var}(\varphi_{IF}) = \mathsf{Var}(\varphi_{EIF}) + \mathsf{Var}\left(\Pi\{\varphi_{IF} \mid \dot{\mathcal{P}}_{\text{ind}}^{\perp}\}\right) \geq \mathsf{Var}(\varphi_{EIF})$.

3.3. Estimators leveraging additional structure

We briefly discuss the possibility of further improving the efficiency of φ_{EIF} in (10) by leveraging additional structures of $\mathcal{P}$. If we further include the additive and independent noise assumption (2), known as the location-shift regression model (Tsiatis, 2006), φ_{EIF} can be refined accordingly and the improved influence function requires the distribution of the error term E_Y (in (2)). However, E_Y is unobserved, and its distribution is difficult to estimate with sufficient accuracy to ensure the asymptotic distribution in Section 4. Therefore, the efficiency gain from incorporating (2) is primarily conceptual rather than practical, and we therefore continue with φ_{EIF} in what follows (see Chapter 5 of Tsiatis (2006) for a more detailed discussion).

If $\mathbb{P}_{\boldsymbol{W}}$ is known, φ_{EIF} can be further improved, and we provide the derivation of this efficient influence function in Corollary 3. However, in many real-data scenarios, including the application in Section 6, the distribution $\mathbb{P}_{\boldsymbol{W}}$ is unavailable, and thus we regard φ_{EIF} as more relevant and useful.

For cross-fitting, let $L \geq 2$ be the number of folds and let $\{\mathcal{I}_l\}_{l=1}^{L}$ partition $\{1, \ldots, n\}$. Write $n_l = |\mathcal{I}_l|$ for the fold sizes and $\mathcal{I}_{-l} = \{1, \ldots, n\} \setminus \mathcal{I}_l$ for the training indices.

The functions in (9) and (10) refer to a regression-square component $\mathbb{E}[\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]^2]/\mathsf{Var}(Y)$. For a total or interaction explainability ξ, its influence functions $\varphi_{\text{IF}}^{\xi}$ and $\varphi_{\text{EIF}}^{\xi}$ are the corresponding linear combinations specified by (4) and (5). In the inference statements and algorithms, φ_{IF} and φ_{EIF} denote these full-target influence functions, with the superscript ξ omitted. In these inference statements and algorithms, $\sigma^2 = \mathsf{Var}\{\varphi(Y, \boldsymbol{W}; \mathbb{P})\}$ denotes the variance of the chosen influence function.

4. INFERENCE

4.1. Asymptotic distribution of one-step correction estimator

Write $\widehat{\varphi}_{\text{IF}} = \varphi_{\text{IF}}(Y, \boldsymbol{W}; \widehat{\mathbb{P}})$ for the influence function with fitted nuisance components, and similarly for $\widehat{\varphi}_{\text{EIF}}$.

Throughout these asymptotic statements, K and the number of folds L are fixed, $\min_l n_l/n$ is bounded away from zero, and the partition is chosen independently of the observations. Each nuisance estimator is measurable with respect to its training fold and any fitting randomness independent of the held-out observations. Assume $\mathsf{Var}(Y) > 0$. Unqualified norms are $L_2(\mathbb{P})$ norms, with functions of a subvector lifted to the full observation; marginal-density norms are $L_2(\mathbb{P}_{W_k})$ norms. For probability laws, write $\|Q - P\|_{\text{var}} := \int |dQ - dP|$. All conditions below hold for each fold and each of the finitely many regression-square components entering the estimand. Write $\widehat{\mu}_{-\mathcal{S},-l}$ for the fitted reduced conditional mean $\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]$. Use the training-sample moments for φ_{IF} and the fitted integrated moments for φ_{EIF} specified in Algorithm 1 and table 4.

Proposition 3 (One-step correction estimator based on $\varphi_{\mathbf{IF}}$) *We assume the following.*

(C1) (Boundedness) There exists $C < \infty$ such that $|Y|, |\mu|, |\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]|, |\widehat{\mu}_{-l}|, |\widehat{\mu}_{-\mathcal{S},-l}| \le C$ almost surely. The regression bounds hold on the supports of both the true and fitted laws.

(C2) (Consistency) The regression errors $\|\widehat{\mu}_{-l} - \mu\|$, $\|\widehat{\mu}_{-\mathcal{S},-l} - \mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]\|$ and $\|\widehat{\mathbb{P}}_{\boldsymbol{W},-l} - \mathbb{P}_{\boldsymbol{W}}\|_{\mathrm{var}}$ converge to zero in probability.

(C3) (Rates) These three errors are $o_{\mathbb{P}}(n^{-1/4})$.

Under C1, the empirical training mean and variance automatically have $O_{\mathbb{P}}(n^{-1/2}) = o_{\mathbb{P}}(n^{-1/4})$ errors. Then the estimator in Algorithm 1 satisfies

$$\sqrt{n}(\widehat{\xi}_{IF} - \xi) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \varphi_{IF}(Y_i, \boldsymbol{W}_i; \mathbb{P}) + o_{\mathbb{P}}(1) \xrightarrow{d} \mathcal{N}(0, \mathsf{Var}\{\varphi_{IF}(Y, \boldsymbol{W}; \mathbb{P})\}) .$$

Thus it is asymptotically linear at $\mathbb{P}$.

Proposition 4 (One-step correction estimator based on $\varphi_{\mathbf{EIF}}$) *Assume Assumption 1, the preceding sampling conventions, and C1 for $Y, \mu, \widehat{\mu}_{-l}$. Replace the joint-law condition by $\|\widehat{p}_{W_k,-l} - p_{W_k}\|_{L_2(\mathbb{P}_{W_k})} = o_{\mathbb{P}}(n^{-1/4})$ for every k, and assume $\|\widehat{\mu}_{-l} - \mu\| = o_{\mathbb{P}}(n^{-1/4})$ and $\|\widehat{v}_{-l} - v\| = o_{\mathbb{P}}(n^{-1/4})$. Use the fitted product law and the reduced regression defined in Table 4. Further assume*

(C4) (Positivity) For each coordinate, the true and fitted marginal densities are supported on the same fixed support and have a common dominating measure. For some $\underline{c} > 0$, $p_{W_k}, \widehat{p}_{W_k,-l} \ge \underline{c}$ almost everywhere on that coordinate's support, for every k, l.

(C5) (Donsker class) There is a fixed $\mathbb{P}$-Donsker class $\mathcal{F}$ such that, for each k, l, $\mathbb{P}\{(p_{W_k} - \widehat{p}_{W_k,-l})^2 \in \mathcal{F}\} \to 1$.

For this efficient construction, the reduced-regression and coherent fitted-variance quarter rates follow from these assumptions and need not be imposed separately. Then

$$\sqrt{n}(\widehat{\xi}_{EIF} - \xi) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \varphi_{EIF}(Y_i, \boldsymbol{W}_i; \mathbb{P}) + o_{\mathbb{P}}(1) \xrightarrow{d} \mathcal{N}(0, \mathsf{Var}\{\varphi_{EIF}(Y, \boldsymbol{W}; \mathbb{P})\}) .$$

Proofs of Propositions 3 and 4 are provided in Appendix D.

4.2. Non-degenerate null

For $\mathsf{Var}(\varphi) > 0$, where φ may denote either φ_{IF} or φ_{EIF}, Propositions 3 and 4 imply that we can construct the confidence interval for any significance level $\alpha \in (0, 1)$ as

$$\Big[\, \widehat{\xi} - \Phi^{-1}(1 - \alpha/2) \frac{\widehat{\sigma}}{\sqrt{n}},\ \widehat{\xi} + \Phi^{-1}(1 - \alpha/2) \frac{\widehat{\sigma}}{\sqrt{n}} \,\Big]. \tag{12}$$

Here Φ denotes the standard normal distribution function, and $\widehat{\sigma}$ is the square root of the consistent plug-in variance estimator in Algorithm 1.

Corollary 2 *Under the assumptions of Proposition 4, if $\mathsf{Var}\{\varphi_{EIF}(Y, \boldsymbol{W}; \mathbb{P})\} > 0$, the variance estimator in Algorithm 1 is consistent for this variance, and the confidence interval in (12) has coverage tending to $1 - \alpha$. The same conclusion holds for $\widehat{\xi}_{IF}$ under Proposition 3 and positive variance of its true influence function.*

4.3. Degenerate null

When $\xi(\vee_{k\in\mathcal{S}} W_k) = 0$ for some $\mathcal{S} \subseteq [K]$, the leading term in the expansion of the estimation error vanishes (Propositions 3 and 4), and the asymptotic distribution cannot be directly used to construct confidence interval[2]. Here, we exploit the structure of the null and the independence (1) to introduce a randomization-based inference procedure. Compared with existing approaches to null degeneracy (Section 2.2), our method is finite-sample valid and avoids sample splitting or added noise (and preventing the associated loss of power). As the first step, we establish an equivalent statement of the zero total explainability null.

Proposition 5 *Assume Assumption 1, the null $\xi(\vee_{k\in\mathcal{S}} W_k) = 0$ is equivalent to*

$$Y(w_{\mathcal{S}}, w_{-\mathcal{S}}) = Y(w'_{\mathcal{S}}, w_{-\mathcal{S}}),$$

almost surely over potential outcomes and assignment pairs $(\boldsymbol{W}_{\mathcal{S}}, \boldsymbol{W}'_{\mathcal{S}}, \boldsymbol{W}_{-\mathcal{S}})$. For discrete assignments, this holds almost surely over potential outcomes at each pair with positive assignment probability.

Proof of Proposition 5 is provided in Appendix D. Proposition 5 implies that, holding components $W_{-\mathcal{S}}$ fixed, changing $W_{\mathcal{S}}$ does not affect the outcome Y. In finite samples, this yields, almost surely under supported assignments, the partially sharp null (Fisher, 1918; Zhang and Zhao, 2023),

$$Y_i(w_{\mathcal{S}}, w_{-\mathcal{S}}) = Y_i(w'_{\mathcal{S}}, w_{-\mathcal{S}}), \quad \forall i \in [n]. \tag{13}$$

Under the null (13), for reassignments $W^*_{\mathcal{S},i}$ drawn from the specified assignment law, we can impute the corresponding potential outcome almost surely as $Y_i\big(W^*_{\mathcal{S},i}, W_{-\mathcal{S},i}\big) = Y_i$.

Randomization tests (Section 2.3) are standard for the partially sharp null (13). By Proposition 5, we propose to adopt randomization tests for degenerate nulls. Explicitly, we choose a computable test statistic that tends to be large under the alternative $\xi(\vee_{k\in\mathcal{S}} W_k) > 0$, such as an estimator of $\xi(\vee_{k\in\mathcal{S}} W_k)$. We then generate $W^*_{\mathcal{S},i}$ from the same distribution as $W_{\mathcal{S},i}$ conditional on $W_{-\mathcal{S},i}$, impute the potential outcomes using (13), and recompute the test statistic based on $W^*_{\mathcal{S},i}$, the imputed outcomes, and compare the observed test statistic to those using $W^*_{\mathcal{S},i}$ to compute the p-value.

Note that standard randomization tests require knowing the distribution of $\boldsymbol{W}$ in order to generate $W^*_{\mathcal{S},i}$. However, the joint distribution of $\boldsymbol{W}$ may not be available in our setting. Fortunately, under the independence assumption (1) and the i.i.d. nature of the observations, we have

$$\big(\boldsymbol{W}_{\mathcal{S},\rho(i)}, \boldsymbol{W}_{-\mathcal{S},i}\big)_{i=1}^n \overset{d}{=} \big(\boldsymbol{W}_{\mathcal{S},i}, \boldsymbol{W}_{-\mathcal{S},i}\big)_{i=1}^n, \tag{14}$$

where ρ denotes a permutation of the indices $i \in [n]$. For the randomization test, draw permutations independently and uniformly from the full permutation group, including the identity, and permute $\boldsymbol{W}_{\mathcal{S},i}$ while keeping $\boldsymbol{W}_{-\mathcal{S},i}$ fixed. Apply the same statistic and fitting rule to every dataset, with any auxiliary randomness fixed independently of assignments. The weak inequality in the p-value treats

2. For interaction nulls, zero interaction explainability need not imply a zero unrestricted-model influence function φ_{IF}. Normal inference applies only when the chosen influence function has positive variance and the corresponding regularity conditions hold; the efficient influence function can still be degenerate. Example: consider independent $W_1, W_2, E_Y \sim \mathcal{N}(0,1)$ and $Y = W_1 + W_2 + E_Y$, we have $\xi(W_1), \xi(W_2), \xi(W_1 \vee W_2) > 0$ while $\xi(W_1 \wedge W_2) = 0$. Then $\varphi_{\text{IF}} = -2W_1W_2/\mathsf{Var}(Y)$ for $\xi(W_1 \wedge W_2)$, which does not vanish.

ties conservatively. We summarize the procedure in Algorithm 2. The result below establishes the validity of the randomization test, and the proof is provided in Appendix D. We discuss extensions to other treatment assignment mechanisms, such as blocked design, in Appendix A.4.

Proposition 6 *Assume Assumption 1, under the null $\xi(\vee_{k\in\mathcal{S}} W_k) = 0$, the randomization test in Algorithm 2 controls type I error.*

4.4. Sequential procedure adapting to null degeneracy

Since it is unknown a priori whether $\xi(\vee_{k\in\mathcal{S}} W_k) = 0$, a procedure is needed to distinguish between the degenerate and non-degenerate cases in order to apply the suitable inference procedure above. Explicitly, we propose to use the following sequential approach.

1. Run the randomization test Algorithm 2 for the degenerate null as in Section 4.3 at level α. If the null is not rejected, output $\{0\}$; otherwise, proceed to Step 2.

2. Compute Eq. (12) in Section 4.2 to construct a $(1-\alpha)$ confidence interval and output.

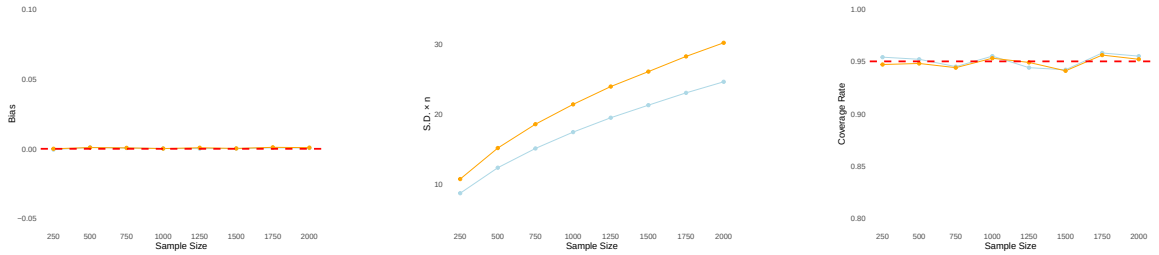
A detailed algorithm of the sequential procedure is provided in Appendix B. We emphasize that although two tests are conducted in the sequential procedure, there is no need to split the significance level α. At the null, noncoverage is contained in first-stage rejection; at a fixed positive alternative, failure to reach the second-stage interval has probability tending to zero. These event bounds, rather than independence of the two stages, justify the following result; its proof is in Appendix D.

Proposition 7 *The sequential procedure has coverage at least $1-\alpha$ at $\xi(\vee_{k\in\mathcal{S}} W_k) = 0$ under Proposition 6. At each fixed $\xi(\vee_{k\in\mathcal{S}} W_k) > 0$, it has asymptotic coverage at least $1-\alpha$ if Corollary 2 applies, including positive influence-function variance, and the randomization test has power tending to one. Replacing the nonrejection output $\{0\}$ by $[0,\infty)$ removes the power requirement; the corresponding second-stage coverage assumptions remain necessary at positive alternatives.*

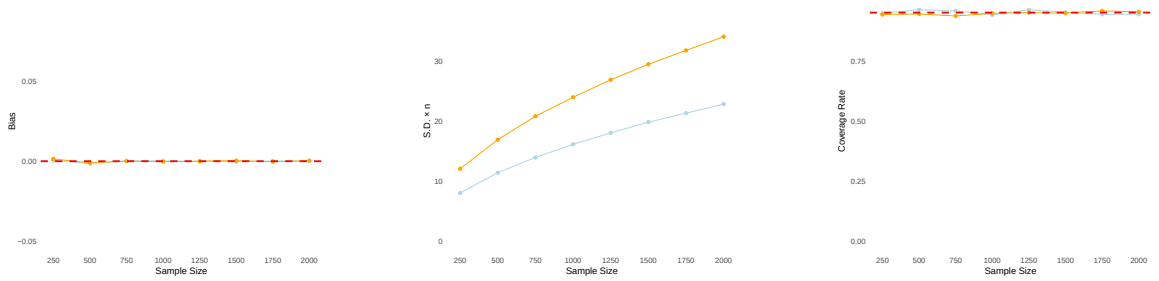
5. SIMULATION

We consider the following data generating mechanism and estimand: independent treatments $W_k \sim \mathcal{N}(0,1)$, $K = 3$, $E_Y \sim \mathcal{N}(0, 0.5)$. The response follows $Y = W_1^3 + W_2^3 + W_3^3 + W_1 W_2^2 + \sigma \cdot W_1^2 W_3 + E_Y$. In these simulations, σ is the interaction coefficient, labelled "noise magnitude" in the plots. The estimands of interest are $\xi(W_3)$, $\xi(W_1 \vee W_3)$ and $\xi(W_1 \wedge W_3)$. We evaluate two different estimators based on influence functions: (1) one-step correction estimator accounting for the independence among $\boldsymbol{W}$ (φ_{EIF}-based estimator); (2) one-step correction estimator ignoring the independence among $\boldsymbol{W}$ (φ_{IF}-based estimator). To start, we implement the two methods using the true nuisance functions.

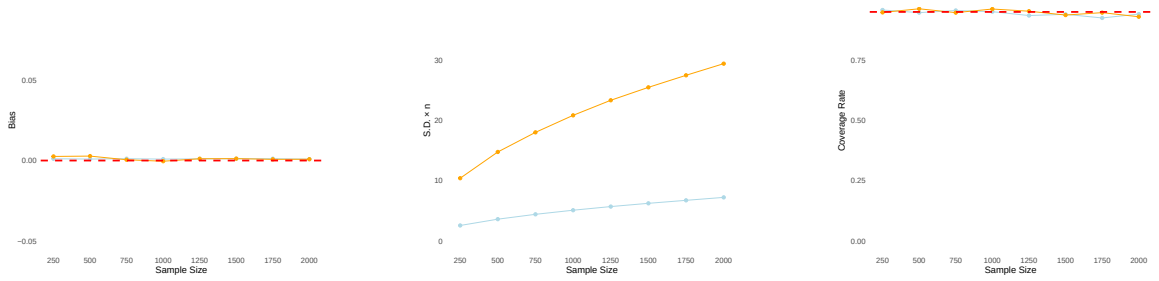
In Figure 1, we vary the sample size. Both methods achieve exact coverage. The φ_{EIF}-based method exhibits lower variance, validating that exploiting independence can improve efficiency. We provide the results with varying σ in Appendix F.1. In Figure 2, we implement the two methods with nuisance functions estimated by the highly adaptive LASSO (HAL) and super learner (package HAL9001 and SUPER LEARNER) (van der Laan et al., 2007; Hejazi et al., 2020). Note that φ_{EIF} involves more nuisance functions than φ_{IF}, for example, φ_{EIF} additionally requires $\mathbb{E}\left[\mathbb{E}[Y \mid \boldsymbol{W}] \cdot \mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}] \mid W_k\right]$ for all $k \in [K]$. With estimated nuisance functions, the φ_{EIF}-based method still exhibits variance reduction, although these gains are less substantial



(a) $\xi(W_3)$; True nuisance functions; Varying sample size; $\sigma = 1$

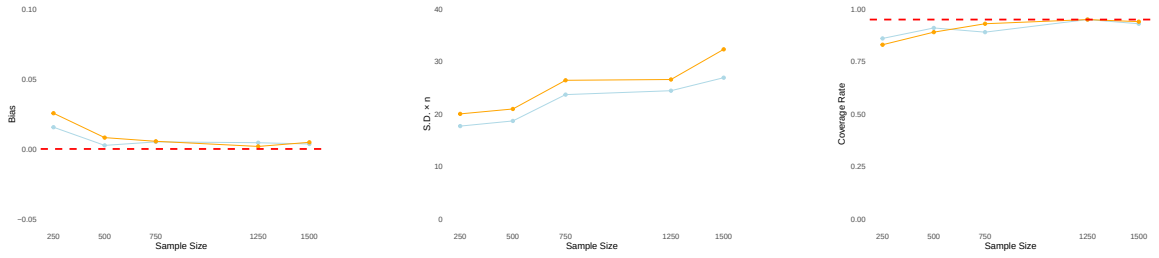


(b) $\xi(W_1 \vee W_3)$; True nuisance functions; Varying sample size; $\sigma = 1$



(c) $\xi(W_1 \wedge W_3)$; True nuisance functions; Varying sample size; $\sigma = 1$

Figure 1: Comparison of bias (left panel), estimated standard deviation times sample size (mid panel), and coverage rate (right panel; significance level $\alpha = 0.05$). φ_{EIF}-based method in blue, φ_{IF}-based method in gold. True nuisance functions are used. Results aggregated over 1000 trials.



$\xi(W_3)$; Estimated nuisance functions; Varying sample size; $\sigma = 1$

Figure 2: Comparison using estimated nuisance functions (details in the caption of Figure 1). Results aggregated over 100 trials.

than those obtained with true nuisance functions. In Appendix F.2, we provide the comparison for interaction explainabilities, where the nuisance estimation is even more complex and the φ_{IF}-based estimator appears competitive. We noticed that the efficient influence function requires more nuisance components, which would result in unstable finite-sample performance (especially under coverage) of the one-step bias-correcting estimator that could offset theoretical efficiency gains. We recommend using the nonparametric influence function (NIF, or IF mentioned in our paper) based one-step bias-correcting estimator in finite samples, especially when the practitioners are not so confident about their estimated nuisances, since it only contains two nuisances. As shown in Appendix F.2, the performance of the IF-based estimator is more robust compared to the φ_{EIF}-based estimator in finite samples using the estimated nuisances. The φ_{EIF}-based estimator is preferred when the nuisances are well-estimated to gain a narrower confidence interval.

6. REAL DATA ANALYSIS

Conjoint analysis is a factorial survey-based experiment in which participants choose between pairs of candidate profiles with several randomly selected characteristics. One of the most cited applications of conjoint analysis is Hainmueller et al. (2015), who examined the role of immigrants' attributes in shaping support for their admission to the United States. In the experiment, 1396 respondents were asked to act as immigration officials and to decide which of a pair of immigrants they would choose for admission. Respondents evaluated a total of five pairings. The attributes were then randomly chosen to form immigrant profiles. We apply the explainability metric and the associated estimation and inference tools proposed to this real dataset. We use the φ_{IF}-based estimator as it requires a simpler set of nuisance functions to estimate[3]. For the nulls of zero total explainability, we do not rule out the degenerate null issue, and thus apply the sequential procedure in Section 4.4. Since the data have a cluster structure, we treat the respondent as the independent unit; in this analysis, n is the number of respondents, m_i is the number of rows for respondent i, and $N = \sum_{i=1}^{n} m_i$ is the total number of rows. We assume respondents are i.i.d. and allow arbitrary within-respondent dependence. We compute fitted row-level influence-function contributions ϕ_{ir} (with cross-fitting done by respondent folds), and aggregate to respondent-level scores $U_i = \sum_{r=1}^{m_i} \phi_{ir}$. The cluster-robust variance for $\widehat{\xi} = \xi(\widehat{\mathbb{P}}) + N^{-1} \sum_{i,r} \phi_{ir}$ is then $\widehat{\text{Var}}(\widehat{\xi}) = \frac{1}{N^2} \cdot \frac{n}{n-1} \sum_{i=1}^{n} (U_i - \overline{U})^2$, $\overline{U} = \frac{1}{n} \sum_{i=1}^{n} U_i$, and we use the $1 - \alpha/2$ quantile $t_{n-1,\,1-\alpha/2}$ of Student's t distribution with $n - 1$ degrees of freedom for confidence intervals. Conditions for the asymptotic approximation and for finite-sample randomization validity, together with the resampling procedure, are given in Appendix G. Table 1 shows that the total explainabilities of *Language*, *Job Experience*, *Job Plan*, *Prior Trips to the U.S.*, *Education & Profession*, and *Origin & Application* reason are all significantly positive, except *Gender*. Among these factors, *Job Plan*, *Education & Profession* exhibit the two highest explainabilities, suggesting an encouraging pattern that immigrants' skills and qualifications are seriously evaluated. None of the reported interaction-explainability intervals in Table 2 excludes zero, including that for *Job Plan* and *Job Experience*. The separate CRT result in Appendix G is evidence under that test's null and should not be read as significance of the explainability interval.

3. Its interaction influence function need not vanish at the null, but normal inference still requires positive influence-function variance.

Table 1: Total explainability

Variable	Total explainability	95% CI
Gender	0.0012	$[-0.0003, 0.0026]$
Language	0.0319	$[0.0240, 0.0398]$
Job experience	0.0164	$[0.0109, 0.0219]$
Job plans	0.0884	$[0.0751, 0.1016]$
Prior trips to U.S.	0.0366	$[0.0275, 0.0457]$
Education & Profession	0.0579	$[0.0471, 0.0687]$
Origin & Application reason	0.0155	$[0.0101, 0.0210]$

Note: The permutation test for Gender returns a p-value of 1, failing to reject the null hypothesis. Its displayed Wald interval is supplementary; the sequential procedure returns $\{0\}$ on nonrejection (or $[0, \infty)$ under the fallback).

Table 2: Interaction explainability

Factor 1	Factor 2	Interaction explainability	95% CI
Job plan	Prior trips to U.S.	-0.0002	$[-0.0032, 0.0029]$
Job plan	Job experience	0.0014	$[-0.0006, 0.0034]$
Job plan	Language	0.0010	$[-0.0017, 0.0038]$

7. DISCUSSION

Using semi-parametric efficiency theory, we develop one-step corrected estimators for causal ANOVA quantities, both with and without incorporating independence across factors. The estimator that incorporates this structure achieves no larger asymptotic variance theoretically and when nuisance components can be accurately estimated, whereas the estimator that does not explicitly leverage independence has a simpler form and may exhibit greater robustness in small samples. To address the issue of null degeneracy, we develop a randomization-based inference procedure and integrate it with the semi-parametric analysis. Applied to a real immigration dataset, our methods identify multiple factors with nonzero explainability. The reported interval for the interaction between *Job Plan* and *Job Experience* includes zero; the separate CRT comparison in Appendix G reports evidence under its own null.

We outline a few future directions. When testing a set of factors and their interactions for nonzero explainability simultaneously, multiplicity arises, and it is of interest to explore proper adjustment. Causal ANOVA quantities exhibit a hierarchical structure (for example, if the total explainability of a factor is zero, then any interaction involving this factor must be zero), which can be exploited via closed testing or gatekeeping to improve power while maintaining error control. In addition, in sequential decision problems (Markov decision processes), the Markov property yields a natural conditional independence. A direction for future research is to investigate the influence functions incorporating this conditional independence to reduce estimator's asymptotic variance. Moreover, when factors are dependent, Causal ANOVA quantities are generally not point-identified. A natural direction is to characterize the identified set and extend the semi-parametric analysis to the estimation and inference for this set.

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mathematical arguments, and assistance with proof formalization. All mathematical results, proofs, interpretations, and cited references were independently checked and verified by the authors, who take full responsibility for the contents of the manuscript.

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Appendix for "Estimation and Inference for Causal Explainability"

Organization. In Appendix A.1, we provide additional definitions of causal ANOVA quantities and discuss their hierarchical structure. In Appendix A.2, we discuss extensions of the NPSEM-IE model (2). In Appendix A.3, we extend the independence condition (1) to conditional independence. In Appendix A.4, we discuss extensions of standard randomization tests to various treatment assignment mechanisms. In Appendix B, we present additional algorithms. In Appendix C, we collect supplementary lemmas and corollaries. In Appendix D, we provide detailed proofs. In Appendix E, we give additional tables. In Appendix F, we report additional empirical studies.

Appendix A. Method extension

A.1. Interaction explainability

We define the interaction effect of a set of treatments iteratively.

Definition 3 (Interaction effect) *For* $\boldsymbol{w}, \boldsymbol{w}' \in \mathbb{R}^K$, *let* $I_{\emptyset,\boldsymbol{w}'}(\boldsymbol{w}) := Y(\boldsymbol{w}')$, $I_{k,\boldsymbol{w}'}(\boldsymbol{w}) := Y(\boldsymbol{w}_k, \boldsymbol{w}'_{-k}) - Y(\boldsymbol{w}')$ *for* $k \in [K]$. *For* $\mathcal{S} \subseteq [K]$,

$$I_{\mathcal{S},\boldsymbol{w}'}(\boldsymbol{w}) := Y(\boldsymbol{w}_{\mathcal{S}}, \boldsymbol{w}'_{-\mathcal{S}}) - \sum_{\mathcal{S}' \subsetneq \mathcal{S}} I_{\mathcal{S}',\boldsymbol{w}'}(\boldsymbol{w}).$$

Equivalently, the interaction term can be represented as a linear combination of the potential outcomes evaluated at a combination of $\boldsymbol{w}_{\mathcal{S}'}, \boldsymbol{w}'_{-\mathcal{S}'}$ for $\mathcal{S}' \subseteq \mathcal{S}$ with coefficients in $\{1, -1\}$,

$$I_{\mathcal{S},\boldsymbol{w}'}(\boldsymbol{w}) = \sum_{\mathcal{S}' \subseteq \mathcal{S}} (-1)^{|\mathcal{S}-\mathcal{S}'|} Y(\boldsymbol{w}_{\mathcal{S}'}, \boldsymbol{w}'_{-\mathcal{S}'}).$$

The interaction effect of W_k and $W_{k'}$ takes the form $I_{\{k,k'\},\boldsymbol{w}'}(\boldsymbol{w}) = Y(w_k, w_{k'}, \boldsymbol{w}'_{-\{k,k'\}}) - Y(w_k, \boldsymbol{w}'_{-k}) - Y(w_{k'}, \boldsymbol{w}'_{-k'}) + Y(\boldsymbol{w}')$. For $K = 2$ and binary W_1, W_2, VanderWeele and Tchetgen (2014) uses $I_{\{1,2\},[0,0]}([1, 1])$ as the interactive effect of W_1 and W_2.

The anchored contrast holds treatments outside $\mathcal{S}$ at the common anchor $\boldsymbol{W}'_{-\mathcal{S}}$. Under the product assignment law, replacing this common complement by $\boldsymbol{W}_{-\mathcal{S}}$ gives a local contrast with the same variance, by exchangeability of the independent copies. This variance equivalence permits computing interaction explainability using a copy only of the treatments in $\mathcal{S}$ ("self-sufficiency"); it is not a pointwise identity of contrasts. The common full anchor is retained in (16).

We define the explainability of the interaction of a set of treatments $W_{\mathcal{S}}$, $\mathcal{S} \subseteq [K]$.

Definition 4 *Let* $\boldsymbol{W}'$ *be an independent copy of* $\boldsymbol{W}$. *For* $\mathcal{S} \subseteq [K]$,

$$\xi(\wedge_{k \in \mathcal{S}} W_k) = \frac{\mathsf{Var}\left(I_{\mathcal{S},\boldsymbol{W}'}(\boldsymbol{W})\right)}{2^{|\mathcal{S}|}\mathsf{Var}\left(Y\right)}. \tag{15}$$

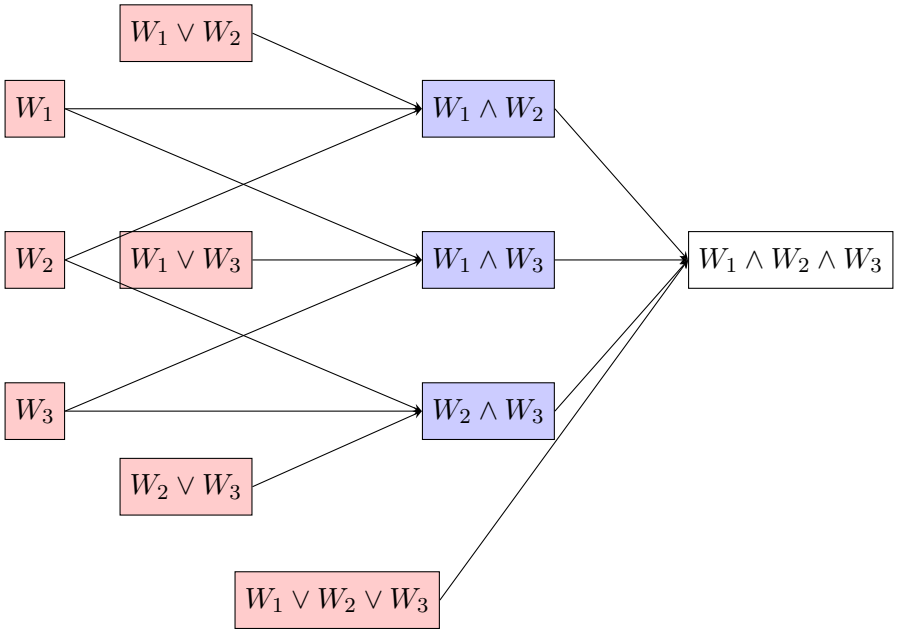


Figure 3: Influence functions of total and interaction explainabilities: zeroth-order, first-order, second-order interaction terms are highlighted in red, blue, and white, respectively. The dependence is derived from the inclusion-exclusion principle and the linearity of influence functions.

The difference $Y - Y(\boldsymbol{W}')$ can be decomposed into the sum of interaction effects with $\mathcal{S} \neq \emptyset$,

$$Y - Y(\boldsymbol{W}') = \sum_{\emptyset \neq \mathcal{S} \subseteq [K]} I_{\mathcal{S},\boldsymbol{W}'}(\boldsymbol{W}). \tag{16}$$

Note that $I_{\mathcal{S},\boldsymbol{W}'}(\boldsymbol{W})$, $I_{\mathcal{S}',\boldsymbol{W}'}(\boldsymbol{W})$ are correlated, therefore $\mathsf{Var}(Y - Y(\boldsymbol{W}'))$ typically does not equal the sum of the variance of $I_{\mathcal{S},\boldsymbol{W}'}(\boldsymbol{W})$. However, we show for independent treatments, $\mathsf{Var}(Y - Y(\boldsymbol{W}'))$ can be represented as a linear combination of the variance of $I_{\mathcal{S},\boldsymbol{W}'}(\boldsymbol{W})$.

Proposition 8 (Anchored variance decomposition) *Under Assumption 1, for every nonempty* $\mathcal{S} \subseteq [K]$,

$$\xi(\vee_{k\in\mathcal{S}} W_k) = \sum_{\emptyset \neq \mathcal{S}' \subseteq \mathcal{S}} (-1)^{|\mathcal{S}'|-1}\, \xi(\wedge_{k\in\mathcal{S}'} W_k), \tag{17}$$

$$\xi(\wedge_{k\in\mathcal{S}} W_k) = \sum_{\emptyset \neq \mathcal{S}' \subseteq \mathcal{S}} (-1)^{|\mathcal{S}'|-1}\, \xi(\vee_{k\in\mathcal{S}'} W_k). \tag{18}$$

Proposition 8 indicates the total explainability and the explainability of interactions admits a relationship resembling the inclusion-exclusion principle in its structural form. Based on it, the influence functions for the explainability of interactions among treatments can be derived from the inclusion-exclusion principle and the linearity property of influence functions (Figure 3).

A.2. Extension of NPSEMs

We next discuss three NPSEM models of increasing generality, with NPSEM-IE (2) as the starting point.

NPSEM-IE. NPSEM-IE is a commonly used framework for causal discovery and inference (Hoyer et al., 2008; Peters et al., 2014). Identification of an unknown causal structure requires additional conditions. Here, the shared-error restriction identifies the causal explainability quantities under the stated assignment assumptions.

NPSEM with additive noise. Consider the NPSEM model where the noise remains additive, but its distribution may depend on the treatment level , i.e., heteroskedastic errors. Additivity and heteroskedasticity alone do not imply co-monotonicity. If the counterfactuals are additionally nondecreasing functions of a common latent rank, the model satisfies the co-monotone property (Gao and Zhao, 2025) under which the joint distribution of all potential outcomes is identifiable. In principle, one could estimate this joint distribution, construct an oracle from it, and sample from this oracle to evaluate the Causal ANOVA quantities. However, learning this joint distribution is statistically challenging, and the inferential properties of such a procedure remain largely unexplored.

General NPSEM. Finally, when the noise is not necessarily additive, the co-monotonicity structure may be violated, and without further restrictions the Causal ANOVA quantities are generally only partially identifiable. Characterizing and conducting inference for the resulting partial identification set is important but beyond the scope of this paper.

As an observable diagnostic, one can compare residual distributions $Y - \mathbb{E}[Y \mid \boldsymbol{W}]$ across treatment levels. Approximate invariance supports the observed additive independent-error restriction, but does not verify the shared-error coupling of potential outcomes. Variation across levels motivates a more general error model.

A.3. Covariates adjustment

If there exists a set of covariates $\boldsymbol{X}$ such that the treatments are unconfounded given $\boldsymbol{X}$, that is $Y(\cdot)_i \perp\!\!\!\perp \boldsymbol{W}_i \mid \boldsymbol{X}_i$, the above analysis can be extended by conditioning on $\boldsymbol{X}$.

Assumption 3 (Unconfounded treatments) *The treatments are independent of the potential outcomes conditional on the covariates,*

$$Y_i(\cdot) \perp\!\!\!\perp \boldsymbol{W}_i \mid \boldsymbol{X}_i.$$

Assumption 4 (Conditionally independent treatments) *Treatments W_k, $k \in [K]$ are mutually independent conditional on the covariates.*

Definition 5 (Total explainability with covariates) *Let $\boldsymbol{W}'$ have the conditional law of $\boldsymbol{W}$ given $\boldsymbol{X}$ and be independent of $(\boldsymbol{W}, Y(\cdot))$ given $\boldsymbol{X}$. For $\mathcal{S} \subseteq [K]$,*

$$\xi(\vee_{k\in\mathcal{S}} W_k) := \frac{\mathbb{E}\left[\mathsf{Var}\left(Y(\boldsymbol{W}) - Y(\boldsymbol{W}'_{\mathcal{S}}, \boldsymbol{W}_{-\mathcal{S}}) \mid \boldsymbol{X}\right)\right]}{2\mathsf{Var}\left(Y(\boldsymbol{W})\right)}. \tag{19}$$

Lemma 1 (Law of total conditional variance) *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $Y \in \mathcal{L}_2(\mathbb{P})$, and let X and W be random elements. Define $\mathcal{G} := \sigma(X)$ and $\mathcal{H} := \mathcal{G} \vee \sigma(W)$. Then, $\mathbb{P}$ a.s.,*

$$\mathsf{Var}(Y \mid \mathcal{G}) = \mathbb{E}[\,\mathsf{Var}(Y \mid \mathcal{H}) \,|\, \mathcal{G}] \;+\; \mathsf{Var}(\mathbb{E}[Y \mid \mathcal{H}] \,|\, \mathcal{G})\,.$$

Equivalently,

$$\mathsf{Var}(Y \mid X) = \mathbb{E}[\,\mathsf{Var}(Y \mid X, W) \,|\, X] \;+\; \mathsf{Var}(\mathbb{E}[Y \mid X, W] \,|\, X) \quad \text{a.s.}$$

The proof comes from the law of total variance.

Lemma 2 *Under Assumption 3 and Assumption 4, additionally assume* $Y(\boldsymbol{w}) = f_Y(\boldsymbol{w}, \boldsymbol{X}) + E_Y$, *with the same error across treatment levels and* $E_Y \perp\!\!\!\perp \boldsymbol{W} \mid \boldsymbol{X}$. *Then Definition 5 admits the equivalent form*

$$\xi(\vee_{k\in\mathcal{S}} W_k) = \frac{\mathbb{E}[\mathbb{E}^2[Y \mid \boldsymbol{W}, \boldsymbol{X}]]}{\mathsf{Var}(Y)} - \frac{\mathbb{E}\left[\mathbb{E}^2[Y \mid \boldsymbol{W}_{-\mathcal{S}}, \boldsymbol{X}]\right]}{\mathsf{Var}(Y)}. \tag{20}$$

Indeed, the shared error cancels from the treatment contrast. Conditional independence of the two draws given $(\boldsymbol{W}_{-\mathcal{S}}, \boldsymbol{X})$ gives $\mathbb{E}[\mathsf{Var}\{Y(\boldsymbol{W}) - Y(\boldsymbol{W}'_{\mathcal{S}}, \boldsymbol{W}_{-\mathcal{S}}) \mid \boldsymbol{X}\}] = 2\mathbb{E}[\mathsf{Var}\{\mathbb{E}[Y \mid \boldsymbol{W}, \boldsymbol{X}] \mid \boldsymbol{W}_{-\mathcal{S}}, \boldsymbol{X}\}]$. Expanding the conditional variance yields the displayed formula.

Definition 6 (Interaction explainability with covariates) *Let* $\boldsymbol{W}'$ *have the conditional law of* $\boldsymbol{W}$ *given* $\boldsymbol{X}$ *and be independent of* $(\boldsymbol{W}, Y(\cdot))$ *given* $\boldsymbol{X}$. *For* $\mathcal{S} \subseteq [K]$,

$$\xi(\wedge_{k\in\mathcal{S}} W_k) = \frac{\mathbb{E}\left[\mathsf{Var}\left(I_{\mathcal{S},\boldsymbol{W}'}(\boldsymbol{W}) \mid \boldsymbol{X}\right)\right]}{2^{|\mathcal{S}|}\mathsf{Var}(Y(\boldsymbol{W}))}. \tag{21}$$

A.4. Extensions of randomization tests

When the assignment mechanism admits additional constraints, we can use a conditional randomization/permutation test that respects the design. The following permutations apply when the conditional assignment law is invariant under the stated transformations; otherwise draws must use that law directly.

1. Under blocking, permutations shall be carried out within each block.
2. Under fixed treatment margins, treatment assignments shall be permuted while keeping the margins fixed.
3. Under repeated measurements, treatment assignments shall be permuted within the measurements of each individual.

The key requirement is joint exchangeability of the observed and all resampled test statistics under the conditioning implied by the design. Independent draws from the same conditional assignment law, or independent uniform draws from an invariant permutation group including the identity, provide this property; pairwise equality in distribution alone is insufficient.

Appendix B. Additional algorithms

4. Interaction inference also requires positive influence-function variance; degeneracy may occur, as in Section 5. Coverage here is for each specified estimand; simultaneous conclusions across factors require multiplicity adjustment.

Algorithm 1: Cross-fitting of one-step correction estimator

Input: Data $\{\boldsymbol{W}_i, Y_i\}_{i=1}^n$; number of folds $L \geq 2$; an influence function φ; nuisance learners for $\mathbb{P}$, $\mu(\boldsymbol{w}) = \mathbb{E}[Y \mid \boldsymbol{W} = \boldsymbol{w}]$, and $\upsilon(\boldsymbol{w}) = \mathbb{E}[Y^2 \mid \boldsymbol{W} = \boldsymbol{w}]$ (needed only for φ_{EIF}).

Partition: Partition the sample into L folds $\{\mathcal{I}_\ell\}_{\ell=1}^L$ with (approximately) equal sizes $n_\ell = |\mathcal{I}_\ell|$.

for $\ell = 1, \ldots, L$ **do**

Nuisance estimation: On $\mathcal{I}_{-\ell}$, obtain nuisance estimators $\widehat{\mu}_{-\ell}(\boldsymbol{w})$ for $\mu(\boldsymbol{w})$, $\widehat{\upsilon}_{-\ell}(\boldsymbol{w})$ for $\upsilon(\boldsymbol{w})$ when using φ_{EIF}, $\widehat{\mathbb{P}}_{W_k,-\ell}$ for $\mathbb{P}_{W_k}$, $k = 1, \ldots, K$, and compute

$$\widehat{\mathbb{P}}_{\boldsymbol{W},-\ell} := \prod_{k=1}^{K} \widehat{\mathbb{P}}_{W_k,-\ell}.$$

For φ_{IF}, use the training mean $\widehat{\mathbb{E}}_{-\ell}[Y] = (n - n_\ell)^{-1} \sum_{i \in \mathcal{I}_{-\ell}} Y_i$ and variance $\mathsf{Var}(Y)_{\widehat{\mathbb{P}}_{-\ell}} = (n - n_\ell)^{-1} \sum_{i \in \mathcal{I}_{-\ell}} (Y_i - \widehat{\mathbb{E}}_{-\ell}[Y])^2$. For φ_{EIF}, use the fitted integrated moments in Table 4. Here $\widehat{\mathbb{P}}_{-\ell}$ denotes the collection of fitted components. Define the estimator arbitrarily if a fitted denominator is zero.

One-step correction: On the held-out fold $\mathcal{I}_\ell$, compute

$$\widehat{\xi}_\ell := \frac{1}{n_\ell} \sum_{i \in \mathcal{I}_\ell} \widehat{\xi}_i, \qquad \widehat{\xi}_i := \xi\big(\widehat{\mathbb{P}}_{-\ell}\big) + \varphi\big(Y_i, \boldsymbol{W}_i; \widehat{\mathbb{P}}_{-\ell}\big).$$

The details of $\xi(\widehat{\mathbb{P}}_{-\ell})$ and $\varphi(Y_i, \boldsymbol{W}_i; \widehat{\mathbb{P}}_{-\ell})$ are given in Table 4.

end

Aggregation: Average the estimates over folds:

$$\widehat{\xi} := \frac{1}{n} \sum_{\ell=1}^{L} n_\ell \widehat{\xi}_\ell.$$

Compute the variance estimator

$$\widehat{\sigma}^2 := \frac{1}{n} \sum_{i=1}^{n} \big(\widehat{\xi}_i - \widehat{\xi}\big)^2.$$

Set $\widehat{\sigma} = \sqrt{\widehat{\sigma}^2}$.

Output: Estimator $\widehat{\xi}$ and its estimated standard error $\widehat{\sigma}/\sqrt{n}$.

Algorithm 2: Randomization test for degenerate null

Input: Data $\{\boldsymbol{W}_i, Y_i\}_{i=1}^n$; test statistic $T(Y, \boldsymbol{W})$ (e.g. an arbitrary estimator of $\xi(\vee_{k\in\mathcal{S}} W_k)$); number of permutations B.

for $b = 1, \ldots, B$ **do**

Draw ρ_b independently and uniformly from all permutations of $[n]$, including the identity; set $\boldsymbol{W}_{\mathcal{S},i}^{(b)} = \boldsymbol{W}_{\mathcal{S},\rho_b(i)}$, keeping $\boldsymbol{W}_{-\mathcal{S},i}$ fixed

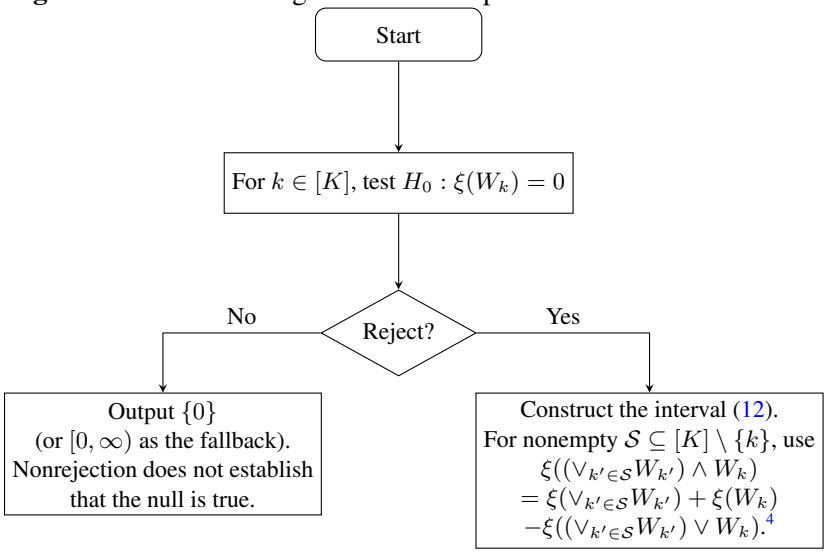
Impute $Y_i(\boldsymbol{W}_i^{(b)}) = Y_i$ and compute $T(Y, \boldsymbol{W}^{(b)})$

end

Compute $P := \dfrac{1}{B+1}\left[1 + \sum_{b=1}^{B} \mathbb{1}\left\{T(Y, \boldsymbol{W}^{(b)}) \geq T(Y, \boldsymbol{W})\right\}\right]$

Output: p-value P

Algorithm 3: General algorithm for independent treatments



Appendix C. Additional lemmas and corollaries

Lemma 3 (*Theorem 5.6.* in Tsiatis (2006)) *All notation in this lemma is local to the lemma. Here W and Z are generic random elements, and β and η are parameters indexing the marginal model p_Z. If no restrictions are put on the conditional density $p_{W|Z}(w|z)$, where the marginal density of Z is assumed to be from the semi-parametric model $p_Z(z, \beta, \eta)$, then the orthogonal complement of the tangent space $\mathscr{T}^{WZ}$ for the semi-parametric model for the joint distribution of (W, Z) (i.e., $(\mathscr{T}^{WZ})^{\perp}$) is equal to the orthogonal complement of the tangent space $\mathscr{T}^{Z}$ for the semi-parametric model of the marginal distribution for Z alone (i.e., $(\mathscr{T}^{Z})^{\perp}$).*

Lemma 4 *Fix $\mathcal{S} \subseteq [K]$ such that $0 < |-\mathcal{S}| < K$. Assume that each of the two blocks $\boldsymbol{W}_{\mathcal{S}}$ and $\boldsymbol{W}_{-\mathcal{S}}$ admits a centered square-integrable function with positive variance. Consider the semiparametric model $\mathcal{P}_{blk}$ for $(Y, \boldsymbol{W}) = (Y, \boldsymbol{W}_{-\mathcal{S}}, \boldsymbol{W}_{\mathcal{S}})$ defined by*

$$p(y, \boldsymbol{w}_{-\mathcal{S}}, \boldsymbol{w}_{\mathcal{S}}) = p_{Y|\boldsymbol{W}} \cdot p_{\boldsymbol{W}_{-\mathcal{S}}} \cdot p_{\boldsymbol{W}_{\mathcal{S}}},$$

where we put no restrictions on $p_{Y|\boldsymbol{W}}$, $p_{\boldsymbol{W}_{-\mathcal{S}}}$, or $p_{\boldsymbol{W}_{\mathcal{S}}}$ beyond domination and square integrability of scores.

Let $\dot{\mathcal{P}}_{\text{blk}}$ be the tangent space of $\mathcal{P}_{blk}$ at $\mathbb{P}$. Let $\mathcal{P}_{np}(Y, \boldsymbol{W}_{-\mathcal{S}})$ be the unrestricted model for the marginal law of $(Y, \boldsymbol{W}_{-\mathcal{S}})$, with tangent space $\dot{\mathcal{P}}_{Y,\boldsymbol{W}_{-\mathcal{S}}} = L_2^0(\mathbb{P}_{Y,\boldsymbol{W}_{-\mathcal{S}}})$.

Then

$$\dot{\mathcal{P}}^{\perp}_{Y,\boldsymbol{W}_{-\mathcal{S}}} = \{0\}, \text{ but } \dot{\mathcal{P}}^{\perp}_{\text{blk}} \neq \{0\}.$$

Proof Since $\mathcal{P}_{\text{np}}(Y, \boldsymbol{W}_{-\mathcal{S}})$ is unrestricted, its tangent space is $L_2^0(\mathbb{P}_{Y,\boldsymbol{W}_{-\mathcal{S}}})$ and hence $\dot{\mathcal{P}}^{\perp}_{Y,\boldsymbol{W}_{-\mathcal{S}}} = \{0\}$.

Under $\mathcal{P}_{\text{blk}}$, the score contributions from $p_{Y|\boldsymbol{W}}$, $p_{\boldsymbol{W}_{-\mathcal{S}}}$, and $p_{\boldsymbol{W}_{\mathcal{S}}}$ vary independently, so the tangent space decomposes orthogonally as

$$\dot{\mathcal{P}}_{\text{blk}} = \dot{\mathcal{P}}_{Y|\boldsymbol{W}} \ \oplus^{\perp} \ \dot{\mathcal{P}}_{\boldsymbol{W}_{-\mathcal{S}}} \ \oplus^{\perp} \ \dot{\mathcal{P}}_{\boldsymbol{W}_{\mathcal{S}}},$$

where

$$\begin{aligned}
\dot{\mathcal{P}}_{Y|\boldsymbol{W}} &= \{s(Y, \boldsymbol{W}) : \mathbb{E}[s \mid \boldsymbol{W}] = 0\}, \\
\dot{\mathcal{P}}_{\boldsymbol{W}_{-\mathcal{S}}} &= \{s(\boldsymbol{W}_{-\mathcal{S}}) : \mathbb{E}[s] = 0\}, \\
\dot{\mathcal{P}}_{\boldsymbol{W}_{\mathcal{S}}} &= \{s(\boldsymbol{W}_{\mathcal{S}}) : \mathbb{E}[s] = 0\}.
\end{aligned}$$

Hence $h(Y, \boldsymbol{W}) \in \dot{\mathcal{P}}^{\perp}_{\text{blk}} = \dot{\mathcal{P}}^{\perp}_{Y|\boldsymbol{W}} \ \cap \ \dot{\mathcal{P}}^{\perp}_{\boldsymbol{W}_{-\mathcal{S}}} \ \cap \ \dot{\mathcal{P}}^{\perp}_{\boldsymbol{W}_{\mathcal{S}}}$ iff it is orthogonal to each component. First, any $h(\boldsymbol{W})$ is orthogonal to $\dot{\mathcal{P}}_{Y|\boldsymbol{W}}$ because for $s \in \dot{\mathcal{P}}_{Y|\boldsymbol{W}}$,

$$\mathbb{E}[h(\boldsymbol{W})s(Y, \boldsymbol{W})] = \mathbb{E}\{h(\boldsymbol{W})\mathbb{E}[s(Y, \boldsymbol{W}) \mid \boldsymbol{W}]\} = 0.$$

So it suffices to find a nonzero $h(\boldsymbol{W})$ orthogonal to both $\dot{\mathcal{P}}_{\boldsymbol{W}_{-\mathcal{S}}}$ and $\dot{\mathcal{P}}_{\boldsymbol{W}_{\mathcal{S}}}$.

In this proof, choose square-integrable functions $u(\boldsymbol{W}_{\mathcal{S}})$ and $v(\boldsymbol{W}_{-\mathcal{S}})$ such that $\mathbb{E}[u(\boldsymbol{W}_{\mathcal{S}})] = \mathbb{E}[v(\boldsymbol{W}_{-\mathcal{S}})] = 0$ and neither is a.s. zero. Define

$$h(\boldsymbol{W}) := u(\boldsymbol{W}_{\mathcal{S}})\, v(\boldsymbol{W}_{-\mathcal{S}}).$$

Then $h(\boldsymbol{W}) \not\equiv 0$. Moreover, for any mean-zero $s(\boldsymbol{W}_{\mathcal{S}})$, using block independence and $\mathbb{E}[v] = 0$,

$$\mathbb{E}[h(\boldsymbol{W})\, s(\boldsymbol{W}_{\mathcal{S}})] = \mathbb{E}[v(\boldsymbol{W}_{-\mathcal{S}})] \ \mathbb{E}[u(\boldsymbol{W}_{\mathcal{S}})s(\boldsymbol{W}_{\mathcal{S}})] = 0,$$

and similarly for any mean-zero $s(\boldsymbol{W}_{-\mathcal{S}})$,

$$\mathbb{E}[h(\boldsymbol{W})\, s(\boldsymbol{W}_{-\mathcal{S}})] = \mathbb{E}[u(\boldsymbol{W}_{\mathcal{S}})]\ \mathbb{E}[v(\boldsymbol{W}_{-\mathcal{S}})s(\boldsymbol{W}_{-\mathcal{S}})] = 0.$$

Thus $h \in \dot{\mathcal{P}}_{\text{blk}}^{\perp}$ and the orthocomplement is nontrivial. ■

Lemma 5 *Assume $K \geq 2$ and the full-independence model $\mathcal{P}_{ind}$ for $(Y, \boldsymbol{W})$:*

$$p(y, \boldsymbol{w}) = p_{Y|\boldsymbol{W}} \prod_{k=1}^{K} p_{W_k},$$

with no restrictions on $p_{Y|\boldsymbol{W}}$ and each p_{W_k} beyond domination and square-integrable scores. Assume that at least two distinct coordinates admit centered square-integrable functions with positive variance. Let $\dot{\mathcal{P}}_{ind}$ be its tangent space at $\mathbb{P}$. Let $\mathcal{P}_{np}(Y)$ be the unrestricted model for the marginal law of Y alone, with tangent space $\dot{\mathcal{P}}_Y = L_2^0(\mathbb{P}_Y)$.

Then

$$\dot{\mathcal{P}}_Y^{\perp} = \{0\},\ \textit{but}\ \dot{\mathcal{P}}_{ind}^{\perp} \neq \{0\}.$$

Proof Since $\mathcal{P}_{\text{np}}(Y)$ is unrestricted, $\dot{\mathcal{P}}_Y = L_2^0(\mathbb{P}_Y)$ and thus $\dot{\mathcal{P}}_Y^{\perp} = \{0\}$.

Under $\mathcal{P}_{\text{ind}}$, the tangent space decomposes orthogonally as

$$\dot{\mathcal{P}}_{\text{ind}} = \bigoplus_{k\in[K]}^{\perp} \dot{\mathcal{P}}_{W_k} \oplus^{\perp} \dot{\mathcal{P}}_{Y|\boldsymbol{W}},$$

where $\dot{\mathcal{P}}_{Y|\boldsymbol{W}} = \{s(Y, \boldsymbol{W}) : \mathbb{E}[s \mid \boldsymbol{W}] = 0\}$ and $\dot{\mathcal{P}}_{W_k} = \{s(W_k) : \mathbb{E}[s] = 0\}$.

In this proof, fix distinct indices $i \neq j$ and choose square-integrable $u(W_i), v(W_j)$ with

$$\mathbb{E}[u(W_i)] = \mathbb{E}[v(W_j)] = 0$$

and not a.s. zero. Let

$$h(\boldsymbol{W}) := u(W_i)v(W_j).$$

Then $h(\boldsymbol{W}) \not\equiv 0$. As in Lemma 4, $h(\boldsymbol{W}) \perp \dot{\mathcal{P}}_{Y|\boldsymbol{W}}$. For any $s(W_k) \in \dot{\mathcal{P}}_{W_k}$, mutual independence and centering imply

$$\mathbb{E}[h(\boldsymbol{W})s(W_i)] = \mathbb{E}[v(W_j)]\,\mathbb{E}[u(W_i)s(W_i)] = 0,$$
$$\mathbb{E}[h(\boldsymbol{W})s(W_j)] = \mathbb{E}[u(W_i)]\,\mathbb{E}[v(W_j)s(W_j)] = 0,$$

and for $k \notin \{i, j\}$,

$$\mathbb{E}[h(\boldsymbol{W})s(W_k)] = \mathbb{E}[h(\boldsymbol{W})]\,\mathbb{E}[s(W_k)] = 0.$$

Thus $h \in \dot{\mathcal{P}}_{\text{ind}}^{\perp} = \bigcap_{k\in[K]} \dot{\mathcal{P}}_{W_k}^{\perp} \cap \dot{\mathcal{P}}_{Y|\boldsymbol{W}}^{\perp}$, proving the orthocomplement is nontrivial. ■

Corollary 3 *Under the moment and positive-variance conditions of Proposition 2, if $\mathbb{P}_{\boldsymbol{W}}$ is known and $\mathbb{P}_{Y|\boldsymbol{W}}$ is unrestricted, the efficient influence function takes the form*

$$\frac{2\,(Y-\mathbb{E}[Y\mid \boldsymbol{W}])\,\mathbb{E}[Y\mid \boldsymbol{W}_{-\mathcal{S}}]}{\mathsf{Var}\,(Y)} - \frac{\left\{\,(Y-\mathbb{E}[Y])^2-\mathbb{E}\left[(Y-\mathbb{E}[Y])^2\mid \boldsymbol{W}\right]\right\}\cdot\mathbb{E}\left[\mathbb{E}\left[Y\mid \boldsymbol{W}_{-\mathcal{S}}\right]^2\right]}{\mathsf{Var}\,(Y)^2}.$$

Proof With $\mathbb{P}_{\boldsymbol{W}}$ fixed, the tangent space is $\{s\in L_2^0(\mathbb{P}):\mathbb{E}[s\mid \boldsymbol{W}]=0\}$. The efficient influence function is therefore the unrestricted influence function in (9) minus its conditional mean given $\boldsymbol{W}$:

$$\varphi_{\text{EIF}}=\varphi_{\text{IF}}-\mathbb{E}[\varphi_{\text{IF}}\mid \boldsymbol{W}].$$

Substitution of (9) gives the displayed expression. ■

Lemma 6 (*Lemma 2.* in Kennedy et al. (2020)) *Let $\widehat{f}(\mathbf{o})$ be a function estimated from a sample $\mathbf{O}^N=(\mathbf{O}_{n+1},\ldots,\mathbf{O}_N)$, and let $\mathbb{P}_n$ denote the empirical measure over $(\mathbf{O}_1,\ldots,\mathbf{O}_n)$, which is independent of $\mathbf{O}^N$. In this lemma, $\mathbb{P}$ denotes the common observation law, $f\in L_2(\mathbb{P})$ is fixed, and the norm is in $L_2(\mathbb{P})$. Then*

$$(\mathbb{P}_n-\mathbb{P})(\widehat{f}-f)=O_{\mathbb{P}}\left(\frac{\|\widehat{f}-f\|}{\sqrt{n}}\right).$$

Lemma 7 *In this lemma, let $\boldsymbol{W}_1,\ldots,\boldsymbol{W}_n$ be i.i.d. from $\mathbb{P}_{\boldsymbol{W}}$, and fix a fold index l. Let $\mathcal{I}_{-l}=[n]\backslash\mathcal{I}_l$ be its training indices, with training size $n_{-l}=n-n_l\to\infty$. Let h_{-l} be training-measurable and suppose $\mathbb{P}\{h_{-l}\in\mathcal{H}\}\to 1$ for a fixed class $\mathcal{H}$ whose square class $\mathcal{F}=\{h^2:h\in\mathcal{H}\}$ is $\mathbb{P}_{\boldsymbol{W}}$-Donsker. Then*

$$\left|\frac{1}{n_{-l}}\sum_{i\in\mathcal{I}_{-l}}h_{-l}(\boldsymbol{W}_i)^2-\|h_{-l}\|^2_{L_2(\mathbb{P}_{\boldsymbol{W}})}\right|=O_{\mathbb{P}}(n_{-l}^{-1/2}).$$

If additionally $\|h_{-l}\|_{L_2(\mathbb{P}_{\boldsymbol{W}})}\to_{\mathbb{P}}0$, the bound improves to $o_{\mathbb{P}}(n_{-l}^{-1/2})$. For fixed proportional folds this is $o_{\mathbb{P}}(n^{-1/2})$.

Proof The difference is the absolute centered training average of h_{-l}^2. Donsker tightness gives the stated big-O bound uniformly over $\mathcal{F}$. For the strengthened conclusion, in this proof put $f_{-l}=h_{-l}^2\geq 0$, so $\mathbb{P}_{\boldsymbol{W}}f_{-l}=\|h_{-l}\|_2^2\to_{\mathbb{P}}0$. A Donsker class is totally bounded in the covariance semimetric $\|(f-\mathbb{P}_{\boldsymbol{W}}f)-(g-\mathbb{P}_{\boldsymbol{W}}g)\|_{L_2(\mathbb{P}_{\boldsymbol{W}})}$. On a class of nonnegative functions this implies that, for every $\varepsilon>0$, some $\delta>0$ satisfies $\mathbb{P}_{\boldsymbol{W}}f<\delta\Rightarrow\|f\|_2<\varepsilon$. Indeed, otherwise a sequence with $\mathbb{P}_{\boldsymbol{W}}f_j\to 0$ and $\|f_j\|_2\geq\varepsilon$ has a subsequence whose centered functions converge in L_2. The uncentered functions have the same limit, which is nonnegative and has mean zero, hence is zero, a contradiction. This deterministic implication gives $\|f_{-l}\|_2\to_{\mathbb{P}}0$. Adjoin zero to $\mathcal{F}$ and use asymptotic equicontinuity to obtain $n_{-l}^{-1/2}\sum_{i\in\mathcal{I}_{-l}}\{f_{-l}(\boldsymbol{W}_i)-\mathbb{P}_{\boldsymbol{W}}f_{-l}\}=o_{\mathbb{P}}(1)$. The probability of being outside the class tends to zero and does not change the conclusion. No independence between h_{-l} and its training sample is needed, and no extra L_4 or sup-norm assumption has been added. ■

Appendix D. Omitted proofs

D.1. Proof of Proposition 1

Proof

$$
\begin{aligned}
\xi(\vee_{k\in\mathcal{S}}W_k) :=& \frac{\mathsf{Var}\left(Y - Y(\boldsymbol{W}'_{\mathcal{S}}, \boldsymbol{W}_{-\mathcal{S}})\right)}{2\mathsf{Var}\left(Y\right)} \\
=& \frac{\mathbb{E}\left[\mathsf{Var}\left(Y - Y(\boldsymbol{W}'_{\mathcal{S}}, \boldsymbol{W}_{-\mathcal{S}}) \mid Y(\cdot), \boldsymbol{W}_{-\mathcal{S}}\right)\right]}{2\mathsf{Var}\left(Y\right)} \\
&+ \frac{\mathsf{Var}\left[\mathbb{E}\left(Y - Y(\boldsymbol{W}'_{\mathcal{S}}, \boldsymbol{W}_{-\mathcal{S}}) \mid Y(\cdot), \boldsymbol{W}_{-\mathcal{S}}\right)\right]}{2\mathsf{Var}\left(Y\right)} \text{ (Equation (1))} \\
=& \frac{\mathbb{E}\left[\mathsf{Var}\left(Y \mid Y(\cdot), \boldsymbol{W}_{-\mathcal{S}}\right)\right] + \mathbb{E}\left[\mathsf{Var}\left(Y(\boldsymbol{W}'_{\mathcal{S}}, \boldsymbol{W}_{-\mathcal{S}}) \mid Y(\cdot), \boldsymbol{W}_{-\mathcal{S}}\right)\right]}{2\mathsf{Var}\left(Y\right)} \text{ (Assumption 1)} \\
=& \frac{\mathbb{E}\left[\mathsf{Var}\left(Y \mid Y(\cdot), \boldsymbol{W}_{-\mathcal{S}}\right)\right]}{\mathsf{Var}\left(Y\right)} \text{ (independent copy)} \\
=& \frac{\mathbb{E}[\mathsf{Var}(Y \mid \boldsymbol{W}_{-\mathcal{S}}, E_Y)]}{\mathsf{Var}\left(Y\right)} \quad \text{(Definition 1)} \\
=& \frac{\mathbb{E}[\mathsf{Var}(\mathbb{E}[Y \mid \boldsymbol{W}] + E_Y - \mathbb{E}[E_Y] \mid \boldsymbol{W}_{-\mathcal{S}}, E_Y)]}{\mathsf{Var}\left(Y\right)} \quad \text{(Assumption 2)} \\
=& \frac{\mathbb{E}\left[\mathsf{Var}(\mathbb{E}[Y \mid \boldsymbol{W}] \mid \boldsymbol{W}_{-\mathcal{S}})\right]}{\mathsf{Var}\left(Y\right)} \quad (\boldsymbol{W} \perp\!\!\!\perp E_Y) \\
=& \frac{\mathbb{E}[\mathbb{E}^2\left[Y \mid \boldsymbol{W}\right]]}{\mathsf{Var}\left(Y\right)} - \frac{\mathbb{E}\left[\mathbb{E}^2[Y \mid \boldsymbol{W}_{-\mathcal{S}}]\right]}{\mathsf{Var}\left(Y\right)}. \quad \text{(Definition of variance)}
\end{aligned}
$$

■

D.2. Proof of Proposition 2

Proof

In this proof, we write the parameter of interest as $\Psi := \psi \circ T$, where $\Psi : \mathcal{P} \to \mathbb{R}$, $\psi : \mathbb{D} \to \mathbb{R}$ and $T : \mathcal{P} \to \mathbb{D}$. Let $T_1 := \mathbb{E}\Big[\mathbb{E}\{Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}\}^2\Big]$, $T_2 := \mathsf{Var}(Y)$, and define $T := (T_1, T_2)$, $\psi(x, y) := x/y$. Assume that $T_2 > 0$, so that $T \in \mathbb{D} := \{(x, y) \in \mathbb{R}^2 : y > 0\}$. Since ψ is C^1 on the open set $\mathbb{D}$, it is Fréchet differentiable there, and hence Hadamard differentiable. By *Lemma 1.* in Williamson et al. (2021), T is pathwise differentiable, i.e. Hadamard differentiable along the tangent space $\dot{\mathcal{P}}$ (van der Vaart, 1991; Bickel et al., 1993). Since Hadamard differentiability allows chain rule (Averbukh and Smolyanov, 1967), $\Psi = \psi \circ T$ is Hadamard differentiable along $\dot{\mathcal{P}}$, i.e. pathwise differentiable, with derivative

$$
\dot{\Psi}(S) = \dot{\psi}_T \circ \dot{T} = \frac{\dot{T}_1(S)}{T_2} - \frac{T_1}{T_2^2}\dot{T}_2(S) \in \mathbb{R}, \quad S \in \dot{\mathcal{P}}.
$$

Then by Riesz-Fréchet representation theorem, there exists a unique $\varphi_{\text{EIF}} \in \dot{\mathcal{P}}$ such that

$$
\dot{\Psi}(S) = \langle \varphi_{\text{EIF}}, S\rangle_{L_2} = \langle \frac{\varphi_{T_1,\text{EIF}}}{T_2} - \frac{T_1}{T_2^2}\varphi_{T_2,\text{EIF}}, S\rangle_{L_2}.
$$

If we ignore the independence among treatments, then the model we consider is a non-parametric model $\mathcal{P}_{\text{np}}$ with tangent space $\dot{\mathcal{P}}_{\text{np}} = L_2^0(\mathbb{P})$. Under the assumption of discrete treatments, for $-\mathcal{S} \subseteq [K]$, we use the operator $\mathbb{IF}$ for pathwise differentiable functionals (Kennedy, 2024) to get the non-parametric influence function of the numerator, where, in this discrete calculation, λ is counting measure and $p_{\boldsymbol{W}_{-\mathcal{S}}}$ is the corresponding probability mass function:

$$
\begin{aligned}
\varphi_{T_1,\text{IF}} &= \mathbb{IF}\Big\{ \mathbb{E}\left[\mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}\right]^2 \right] \Big\} \\
&= \mathbb{IF}\Big\{ \int \mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}} = \boldsymbol{w}_{-\mathcal{S}}\right]^2 p_{\boldsymbol{W}_{-\mathcal{S}}}(\boldsymbol{w}_{-\mathcal{S}}) \lambda(d\boldsymbol{w}_{-\mathcal{S}}) \Big\} \\
&= \int \mathbb{IF}\{\mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}} = \boldsymbol{w}_{-\mathcal{S}}\right]^2\} p_{\boldsymbol{W}_{-\mathcal{S}}}(\boldsymbol{w}_{-\mathcal{S}}) \lambda(d\boldsymbol{w}_{-\mathcal{S}}) \\
&\quad + \int \mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}} = \boldsymbol{w}_{-\mathcal{S}}\right]^2 \mathbb{IF}(p_{\boldsymbol{W}_{-\mathcal{S}}}(\boldsymbol{w}_{-\mathcal{S}})) \lambda(d\boldsymbol{w}_{-\mathcal{S}}) \\
&= \int 2\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}} = \boldsymbol{w}_{-\mathcal{S}}] \frac{\mathbb{1}(\boldsymbol{W}_{-\mathcal{S}} = \boldsymbol{w}_{-\mathcal{S}})}{p_{\boldsymbol{W}_{-\mathcal{S}}}(\boldsymbol{w}_{-\mathcal{S}})} \\
&\qquad \cdot (Y(\boldsymbol{W}) - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}} = \boldsymbol{w}_{-\mathcal{S}}])\, p_{\boldsymbol{W}_{-\mathcal{S}}}(\boldsymbol{w}_{-\mathcal{S}}) \lambda(d\boldsymbol{w}_{-\mathcal{S}}) \\
&\quad + \int \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}} = \boldsymbol{w}_{-\mathcal{S}}]^2 \left(\mathbb{1}(\boldsymbol{W}_{-\mathcal{S}} = \boldsymbol{w}_{-\mathcal{S}}) - p_{\boldsymbol{W}_{-\mathcal{S}}}(\boldsymbol{w}_{-\mathcal{S}}) \right) \lambda(d\boldsymbol{w}_{-\mathcal{S}}) \\
&= 2Y(\boldsymbol{W})\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2 - \mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right].
\end{aligned}
$$

The same numerator formula holds without assuming discrete treatments. In this proof, for a bounded mean-zero score s, use the local path $d\mathbb{P}_\varepsilon = (1 + \varepsilon s)\, d\mathbb{P}$. The reduced regression under this path satisfies

$$
\mathbb{E}_{\mathbb{P}_\varepsilon}[Y \mid \boldsymbol{W}_{-\mathcal{S}}] = \frac{\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}] + \varepsilon \mathbb{E}[Ys \mid \boldsymbol{W}_{-\mathcal{S}}]}{1 + \varepsilon \mathbb{E}[s \mid \boldsymbol{W}_{-\mathcal{S}}]}.
$$

Consequently,

$$
\begin{aligned}
T_1(\mathbb{P}_\varepsilon) &= \mathbb{E}\left[\frac{(\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}] + \varepsilon \mathbb{E}[Ys \mid \boldsymbol{W}_{-\mathcal{S}}])^2}{1 + \varepsilon \mathbb{E}[s \mid \boldsymbol{W}_{-\mathcal{S}}]} \right], \\
\frac{d}{d\varepsilon} T_1(\mathbb{P}_\varepsilon)\Big|_{\varepsilon=0} &= \mathbb{E}[2\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]\mathbb{E}[Ys \mid \boldsymbol{W}_{-\mathcal{S}}] - \mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]^2 \mathbb{E}[s \mid \boldsymbol{W}_{-\mathcal{S}}]] \\
&= \mathbb{E}[(2Y\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}] - \mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]^2 - \mathbb{E}[\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]^2])s].
\end{aligned}
$$

For sufficiently small ε, the denominator is bounded away from zero. Boundedness of the score s and $Y \in L_2(\mathbb{P})$ justify differentiation under the integral. Bounded mean-zero scores are dense in $L_2^0(\mathbb{P})$, and the assumed square-integrability of the displayed gradient makes this derivative pairing continuous. Thus the same tangent-space derivative representation applies for general treatment spaces.

For the denominator, we have its non-parametric influence function as

$$
\varphi_{T_2,\text{IF}} = \mathbb{IF}\Big\{ \mathsf{Var}\left(Y(\boldsymbol{W})\right) \Big\} = \left(Y(\boldsymbol{W}) - \mathbb{E}\left(Y(\boldsymbol{W})\right)\right)^2 - \mathsf{Var}\left(Y(\boldsymbol{W})\right).
$$

So the non-parametric influence function of the parameter of interest is

$$\begin{aligned}\varphi_{\text{IF}} &= \frac{2Y(\boldsymbol{W})\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2 - \mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right]}{\mathsf{Var}\left(Y(\boldsymbol{W})\right)} \\ &\quad - \frac{\begin{array}{c}\left(Y(\boldsymbol{W}) - \mathbb{E}\left(Y(\boldsymbol{W})\right)\right)^2 \cdot \mathbb{E}\left[\mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}\right]^2\right] \\ - \mathsf{Var}\left(Y(\boldsymbol{W})\right) \cdot \mathbb{E}\left[\mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}\right]^2\right]\end{array}}{\mathsf{Var}\left(Y(\boldsymbol{W})\right)^2} \\ &= \frac{2Y(\boldsymbol{W})\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2}{\mathsf{Var}\left(Y(\boldsymbol{W})\right)} \\ &\quad - \mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right] \cdot \left\{\frac{Y(\boldsymbol{W}) - \mathbb{E}\left(Y(\boldsymbol{W})\right)}{\mathsf{Var}\left(Y(\boldsymbol{W})\right)}\right\}^2.\end{aligned}$$

Given $-\mathcal{S} \subseteq [K]$ and fully independent treatments, the joint probability measure of observed data can be factorized as $\mathbb{P}_{Y,\boldsymbol{W}} = \mathbb{P}_{Y|\boldsymbol{W}} \cdot \prod_{j \in -\mathcal{S}} \mathbb{P}_{W_j} \cdot \prod_{i \in \mathcal{S}} \mathbb{P}_{W_i}$. Let $\mathcal{P}_{\text{ind}}$ be the set of all regular densities $p_{Y,\boldsymbol{W}}$ such that

$$p_{Y,\boldsymbol{W}} = p_{Y|\boldsymbol{W}} \cdot \prod_{j \in -\mathcal{S}} p_{W_j} \cdot \prod_{i \in \mathcal{S}} p_{W_i}.$$

The reason why we have $\{W_i : i \in \mathcal{S}\}$ here is that, even though they are not needed to define our parameter of interest, fully independent treatments implies that $p_{\boldsymbol{W}_{-\mathcal{S}}|\boldsymbol{W}_{\mathcal{S}}} = p_{\boldsymbol{W}_{-\mathcal{S}}}$, then, when both blocks satisfy the nondegeneracy conditions of Lemma 4, the orthogonal complement of the tangent space $\dot{\mathcal{P}}_{Y,\boldsymbol{W}_{-\mathcal{S}},\boldsymbol{W}_{\mathcal{S}}}$ for the joint distribution of $(Y, \boldsymbol{W}_{-\mathcal{S}}, \boldsymbol{W}_{\mathcal{S}})$, $\dot{\mathcal{P}}^{\perp}_{Y,\boldsymbol{W}_{-\mathcal{S}},\boldsymbol{W}_{\mathcal{S}}}$, is not the same as the orthogonal complement of the tangent space $\dot{\mathcal{P}}_{Y,\boldsymbol{W}_{-\mathcal{S}}}$ for the joint distribution of $(Y, \boldsymbol{W}_{-\mathcal{S}})$, $\dot{\mathcal{P}}^{\perp}_{Y,\boldsymbol{W}_{-\mathcal{S}}}$. Including these auxiliary variables can therefore improve efficiency, with strict improvement only when the discarded gradient component is nonzero. Then for this semi-parametric model, we can project the non-parametric influence function of the numerator onto the corresponding tangent space,

$$\begin{aligned}\dot{\mathcal{P}}_{\text{ind}} &= \bigoplus_{k \in [K]}^{\perp} \dot{\mathcal{P}}_{W_k} \oplus^{\perp} \dot{\mathcal{P}}_{Y|\boldsymbol{W}} \\ &= \bigoplus_{k \in [K]}^{\perp} \{S(W_k) : \mathbb{E}(S) = 0\} \oplus^{\perp} \{S(Y, \boldsymbol{W}) : \mathbb{E}(S \mid \boldsymbol{W}) = 0\},\end{aligned}$$

to get the efficient influence function

$$\begin{aligned}\varphi_{T_1,\text{EIF}} &= \sum_{k \in [K]} \Pi\left(\mathbb{IF}\left\{\mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right]\right\} \mid \dot{\mathcal{P}}_{W_k}\right) \\ &\quad + \Pi\left(\mathbb{IF}\left\{\mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right]\right\} \mid \dot{\mathcal{P}}_{Y|\boldsymbol{W}}\right) \\ &= \sum_{k \in [K]} \mathbb{E}\left(\mathbb{IF}\left\{\mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right]\right\} \mid W_k\right) + \mathbb{IF}\left\{\mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right]\right\}\end{aligned}$$

$$
\begin{aligned}
&- \mathbb{E}\left(\mathbb{IF}\left\{\mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right]\right\} \mid \boldsymbol{W}\right) \\
=&2\left(Y(\boldsymbol{W}) - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]\right) \cdot \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] \\
&+ \sum_{j \in -\mathcal{S}} \left\{\mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2 \mid W_j\right] - \mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right]\right\} \\
&+ 2\sum_{i \in \mathcal{S}} \left\{\mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}] \cdot \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] \mid W_i\right] - \mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right]\right\}.
\end{aligned}
$$

For the denominator, we can also project its non-parametric influence function onto $\dot{\mathcal{P}}_{\text{ind}}$. Because constraining the marginal distribution of $\boldsymbol{W}$ to be factorizable (mutually independent) implicitly restricts the conditional density $p_{\boldsymbol{W}|Y}$ to only those specific forms that preserve this independence after integrating out Y, which violates the condition of Lemma 3. Projecting onto the displayed tangent space gives

$$
\begin{aligned}
\varphi_{T_2,\text{EIF}} = &\left(Y - \mathbb{E}\left[Y\right]\right)^2 - \mathsf{Var}\left(Y \mid \boldsymbol{W}\right) - \left(\mathbb{E}[Y \mid \boldsymbol{W}] - \mathbb{E}[Y]\right)^2 \\
&+ \sum_{k \in [K]} \left\{\mathbb{E}\left[\left(Y - \mathbb{E}[Y]\right)^2 \mid W_k\right] - \mathsf{Var}\left(Y\right)\right\}.
\end{aligned}
$$

So the efficient influence function of the parameter of interest for $\mathcal{P}_{\text{ind}}$ is

$$
\begin{aligned}
\varphi_{\text{EIF}} = &\frac{2\left(Y(\boldsymbol{W}) - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]\right) \cdot \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]}{\mathsf{Var}\left(Y(\boldsymbol{W})\right)} \\
&+ \frac{2\sum_{i \in \mathcal{S}} \left\{\mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}] \cdot \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] \mid W_i\right] - \mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right]\right\}}{\mathsf{Var}\left(Y(\boldsymbol{W})\right)} \\
&+ \frac{\sum_{j \in -\mathcal{S}} \left\{\mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2 \mid W_j\right] - \mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\right]\right\}}{\mathsf{Var}\left(Y(\boldsymbol{W})\right)} \\
&- \frac{\left\{\left(Y - \mathbb{E}\left[Y\right]\right)^2 - \mathsf{Var}\left(Y \mid \boldsymbol{W}\right) - \left(\mathbb{E}[Y \mid \boldsymbol{W}] - \mathbb{E}[Y]\right)^2\right\} \cdot \mathbb{E}\left[\mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}\right]^2\right]}{\mathsf{Var}\left(Y(\boldsymbol{W})\right)^2} \\
&- \frac{\left\{\sum_{k \in [K]} \left\{\mathbb{E}\left[\left(Y - \mathbb{E}[Y]\right)^2 \mid W_k\right] - \mathsf{Var}\left(Y\right)\right\}\right\} \cdot \mathbb{E}\left[\mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}\right]^2\right]}{\mathsf{Var}\left(Y(\boldsymbol{W})\right)^2}.
\end{aligned}
$$

■

D.3. Proof of Proposition 3

Proof Apply Theorem 1 of Williamson et al. (2021) and its cross-fitted formulation, Theorem 2 in Section 3.3 of Williamson et al. (2023). In this proof, take $X = \boldsymbol{W}$; the full and reduced regressions are μ and $\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]$: total explainability is their R^2 contrast, and interaction explainabilities are fixed finite linear combinations of these contrasts. Use the empirical training moments specified above; the direct correction is first-order equivalent to the corrected component ratio by the comparison at the end of the following proof. ■

D.4. Proof of Proposition 4

The quarter rates make the second-order remainder $o_{\mathbb{P}}(n^{-1/2})$; the estimator's error is generally $O_{\mathbb{P}}(n^{-1/2})$ when the limiting variance is positive. The projection comparison is

$$\mathsf{Var}\{\varphi_{\mathrm{EIF}}(Y, \boldsymbol{W}; \mathbb{P})\} \le \mathsf{Var}\{\varphi_{\mathrm{IF}}(Y, \boldsymbol{W}; \mathbb{P})\}$$

for the true influence functions. Proposition 4 uses marginal-density rates in place of a joint-law rate and the conditional-second-moment rate to control the fitted variance. Its proof does not require a uniform bound on $\widehat{v}$.

Proof All auxiliary notation introduced below is local to this proof. In each decomposition below, the labels a, b, and c refer only to the terms displayed in that decomposition. For each fold l and integrable function f of $(Y, \boldsymbol{W})$, define the held-out empirical operator by

$$\mathbb{P}_n^l f := n_l^{-1} \sum_{i \in \mathcal{I}_l} f(Y_i, \boldsymbol{W}_i).$$

Define the remainder term based on probability measures $\overline{\mathbb{P}}$ and $\mathbb{P}$ as $R(\overline{\mathbb{P}}, \mathbb{P}) = \psi_{\overline{\mathbb{P}}} - \psi + \mathbb{P}\{\varphi_{\overline{\mathbb{P}}}\}$, where ψ is any estimand. We begin with the total-explainability numerator $\Theta = \mathbb{E}[\mu(\boldsymbol{W})^2] - \mathbb{E}[\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]^2]$ and denote its efficient influence function by θ. For $l = 1, \ldots, L$,

$$\begin{aligned}
\widehat{\Theta}_l - \Theta &= \Theta_{\widehat{\mathbb{P}}_{-l}} + \mathbb{P}_n^l\{\theta_{\widehat{\mathbb{P}}_{-l}}\} - \Theta \\
&= (\mathbb{P}_n^l - \mathbb{P})\{\theta_{\widehat{\mathbb{P}}_{-l}}\} + R(\widehat{\mathbb{P}}_{-l}, \mathbb{P}) \\
&= (\mathbb{P}_n^l - \mathbb{P})\{\theta\} + (\mathbb{P}_n^l - \mathbb{P})\{\theta_{\widehat{\mathbb{P}}_{-l}} - \theta\} + R(\widehat{\mathbb{P}}_{-l}, \mathbb{P}).
\end{aligned}$$

For a held-out fold, the true-score average has variance $\mathsf{Var}(\theta)/n_l$. Its normal limit follows from the CLT; no coupling to a normal random variable is asserted. By Lemma 6, the centered fitted-score error is $O_{\mathbb{P}}(\|\theta_{\widehat{\mathbb{P}}_{-l}} - \theta\|_2/\sqrt{n_l})$. It therefore suffices to prove $L_2(\mathbb{P})$ consistency of the fitted score. The fitted product law is $\widehat{\mathbb{P}}_{\boldsymbol{W},-l} = \prod_{k=1}^K \widehat{\mathbb{P}}_{W_k,-l}$, and the true product law is $\mathbb{P}_{\boldsymbol{W}} = \prod_{k=1}^K \mathbb{P}_{W_k}$. For functions of $\boldsymbol{W}$, $\widehat{\mathbb{E}}_{-l}$ denotes integration under this fitted product law, and its conditional expectations integrate the other coordinates. This fitted product law is distinct from the empirical training law. For the remainder of this proof, let λ_k be the common dominating measure in C4 and let t_k denote the marginal total-variation error. By C4 and the specified density norms,

$$\begin{aligned}
t_k &:= \int |p_{W_k} - \widehat{p}_{W_k,-l}| \, d\lambda_k \\
&\le \underline{c}^{-1} \|p_{W_k} - \widehat{p}_{W_k,-l}\|_{L_2(\mathbb{P}_{W_k})} \\
&= o_{\mathbb{P}}(n^{-1/4}).
\end{aligned}$$

The exact form of the first half of $\theta_{\widehat{\mathbb{P}}_{-l}} - \theta$ is

$$
\begin{aligned}
&(\theta_{\widehat{\mathbb{P}}_{-l}} - \theta)_{\text{first half}} \\
&= 2Y(\boldsymbol{W})\widehat{\mu}_{-l} - 2\widehat{\mu}^2_{-l} + \sum_{k\in[K]} \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-l} \mid W_k] - K \cdot \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-l}] \\
&- \Big\{ 2Y(\boldsymbol{W})\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}] - 2\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]^2 \\
&\quad + \sum_{k\in[K]} \mathbb{E}\big[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]^2 \mid W_k\big] - K \cdot \mathbb{E}\big[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]^2\big] \Big\} \\
&= \underbrace{2Y(\boldsymbol{W})\widehat{\mu}_{-l} - 2\widehat{\mu}^2_{-l} - 2Y(\boldsymbol{W})\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}] + 2\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]^2}_{:=a} \\
&\quad + \underbrace{\begin{array}{l} \displaystyle\sum_{k\in[K]} \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-l} \mid W_k] - K \cdot \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-l}] \\ \quad \displaystyle - \sum_{k\in[K]} \mathbb{E}\big[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]^2 \mid W_k\big] + K \cdot \mathbb{E}\big[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]^2\big] \end{array}}_{:=b}.
\end{aligned}
$$

Note that, under C1. and C2., we have

$$
\begin{aligned}
&||a|| = ||2Y(\boldsymbol{W})\widehat{\mu}_{-l} - 2\widehat{\mu}^2_{-l} - 2Y(\boldsymbol{W})\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}] + 2\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]^2|| \\
&= ||2Y(\boldsymbol{W}) \cdot (\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]) - 2\,(\widehat{\mu}_{-l} + \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]) \cdot (\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}])\,|| \\
&= ||2\,(Y(\boldsymbol{W}) - \widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]) \cdot (\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}])\,|| \\
&\lesssim ||\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]|| = o_{\mathbb{P}}(1)
\end{aligned}
$$

and

$$
\begin{aligned}
\|b\| &= \left\| \begin{array}{l} \displaystyle\sum_{k\in[K]} \widehat{\mathbb{E}}_{-l}\big[\widehat{\mu}^2_{-l} \mid W_k\big] - K \cdot \widehat{\mathbb{E}}_{-l}\big[\widehat{\mu}^2_{-l}\big] \\ \quad \displaystyle - \sum_{k\in[K]} \mathbb{E}\Big[\mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2 \mid W_k\Big] + K \cdot \mathbb{E}\Big[\mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2\Big] \end{array} \right\| \\
&\leq \sum_{k\in[K]} \Big\| \widehat{\mathbb{E}}_{-l}\big[\widehat{\mu}^2_{-l} \mid W_k\big] - \mathbb{E}\Big[\mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2 \mid W_k\Big] \Big\| \\
&\quad + K \cdot \Big| \widehat{\mathbb{E}}_{-l}\big[\widehat{\mu}^2_{-l}\big] - \mathbb{E}\Big[\mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2\Big] \Big|.
\end{aligned}
$$

For the first term, for each fixed $k \in [K]$, we have

$$
\begin{aligned}
&\widehat{\mathbb{E}}_{-l}\big[\widehat{\mu}^2_{-l} \mid W_k\big] - \mathbb{E}\Big[\mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2 \mid W_k\Big] \\
&\quad = \int \Big(\widehat{\mu}^2_{-l} - \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2\Big) \prod_{k'\neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} \\
&\qquad + \int \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2 \Big(\prod_{k'\neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} - \prod_{k'\neq k} d\mathbb{P}_{W_{k'}} \Big).
\end{aligned}
$$

Hence, by the triangle inequality and the fact that this quantity is W_k-measurable,

$$\begin{aligned}
&\Big\|\widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-l} \mid W_k] - \mathbb{E}\Big[\mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2 \mid W_k\Big]\Big\| \\
&\quad = \Big\|\widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-l} \mid W_k] - \mathbb{E}\Big[\mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2 \mid W_k\Big]\Big\|_{L_2(\mathbb{P}_{W_k})} \\
&\quad \le \left\| \int \Big(\widehat{\mu}^2_{-l} - \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2\Big) \prod_{k' \neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} \right\|_{L_2(\mathbb{P}_{W_k})} \\
&\qquad + \left\| \int \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2 \Big(\prod_{k' \neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} - \prod_{k' \neq k} d\mathbb{P}_{W_{k'}} \Big) \right\|_{L_2(\mathbb{P}_{W_k})}.
\end{aligned}$$

Write $\widehat{\mu}^2_{-l} - \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2 = \Big(\widehat{\mu}_{-l} + \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]\Big)\Big(\widehat{\mu}_{-l} - \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]\Big)$. For the left-hand side term, under C1., by Minkowski's integral inequality,

$$\begin{aligned}
&\left\| \int \Big(\widehat{\mu}^2_{-l} - \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2\Big) \prod_{k' \neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} \right\|_{L_2(\mathbb{P}_{W_k})} \\
&\quad \le \int \Big\| \Big(\widehat{\mu}_{-l} + \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]\Big)\Big(\widehat{\mu}_{-l} - \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]\Big) \Big\|_{L_2(\mathbb{P}_{W_k})} \prod_{k' \neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} \\
&\quad \le 2C \int \Big\| \widehat{\mu}_{-l} - \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big] \Big\|_{L_2(\mathbb{P}_{W_k})} \prod_{k' \neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} \\
&\quad \le 2C \left(\int \Big\| \widehat{\mu}_{-l} - \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big] \Big\|^2_{L_2(\mathbb{P}_{W_k})} \prod_{k' \neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} \right)^{1/2} \\
&\quad = 2C \left(\int \Big(\widehat{\mu}_{-l} - \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]\Big)^2 d\mathbb{P}_{W_k} \prod_{k' \neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} \right)^{1/2}.
\end{aligned}$$

Using $0 \le (\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}])^2 \le (2C)^2$ and the telescoping identity for products, we get

$$\begin{aligned}
&\left| \int \Big(\widehat{\mu}_{-l} - \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]\Big)^2 d\mathbb{P}_{W_k} \prod_{k' \neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} - \int \Big(\widehat{\mu}_{-l} - \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]\Big)^2 d\mathbb{P}_{\boldsymbol{W}} \right| \\
&\quad = \left| \int \Big(\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]\Big)^2 d\mathbb{P}_{W_k} \Big(\prod_{k' \neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} - \prod_{k' \neq k} d\mathbb{P}_{W_{k'}} \Big) \right| \\
&\quad \le \sum_{j \neq k} \int \Big(\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]\Big)^2 d\mathbb{P}_{W_k} \Big| d\widehat{\mathbb{P}}_{W_j,-l} - d\mathbb{P}_{W_j} \Big| \prod_{\substack{r \neq k \\ r<j}} d\widehat{\mathbb{P}}_{W_r,-l} \prod_{\substack{r \neq k \\ r>j}} d\mathbb{P}_{W_r} \\
&\quad \le (2C)^2 \sum_{j \neq k} \int \Big| d\widehat{\mathbb{P}}_{W_j,-l} - d\mathbb{P}_{W_j} \Big| \underbrace{\int d\mathbb{P}_{W_k} \prod_{\substack{r \neq k \\ r<j}} d\widehat{\mathbb{P}}_{W_r,-l} \prod_{\substack{r \neq k \\ r>j}} d\mathbb{P}_{W_r}}_{=1}
\end{aligned}$$

$$= (2C)^2 \sum_{j \neq k} \int \left| d\widehat{\mathbb{P}}_{W_j,-l} - d\mathbb{P}_{W_j} \right|$$
$$= (2C)^2 \sum_{j \neq k} \int \left| \widehat{p}_{W_j,-l} - p_{W_j} \right| d\lambda_j.$$

So under C2., we have

$$\left| \begin{array}{l} \int \Big(\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}] \Big)^2 d\mathbb{P}_{W_k} \prod_{k' \neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} \\ \quad - \int \Big(\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}] \Big)^2 d\mathbb{P}_{\boldsymbol{W}} \end{array} \right| = o_{\mathbb{P}}(1).$$

Now use the inequality for $x, y \geq 0$,

$$\left| \sqrt{x} - \sqrt{y} \right| = \frac{|x - y|}{\sqrt{x} + \sqrt{y}} \leq \sqrt{|x - y|}.$$

Applying this with

$$x = \int \Big(\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}] \Big)^2 d\mathbb{P}_{W_k} \prod_{k' \neq k} d\widehat{\mathbb{P}}_{W_{k'},-l}, \; y = \int \Big(\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}] \Big)^2 d\mathbb{P}_{\boldsymbol{W}},$$

gives

$$\begin{aligned} &\left(\int \Big(\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}] \Big)^2 d\mathbb{P}_{W_k} \prod_{k' \neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} \right)^{1/2} \\ &\quad \leq \left(\int \Big(\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}] \Big)^2 d\mathbb{P}_{\boldsymbol{W}} \right)^{1/2} \\ &\quad + \left| \begin{array}{l} \int \Big(\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}] \Big)^2 d\mathbb{P}_{W_k} \prod_{k' \neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} \\ \quad - \int \Big(\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}] \Big)^2 d\mathbb{P}_{\boldsymbol{W}} \end{array} \right|^{1/2} \\ &\quad = \Big\| \widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}] \Big\|_{L_2(\mathbb{P}_{\boldsymbol{W}})} + o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1). \end{aligned}$$

For the right-hand side term, we have

$$\begin{aligned} &\left\| \int \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2 \Big(\prod_{k' \neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} - \prod_{k' \neq k} d\mathbb{P}_{W_{k'}} \Big) \right\|_{L_2(\mathbb{P}_{W_k})} \\ &\quad \leq \sup_{w_k} \left| \int \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2 \Big(\prod_{k' \neq k} d\widehat{\mathbb{P}}_{W_{k'},-l} - \prod_{k' \neq k} d\mathbb{P}_{W_{k'}} \Big) \right| \\ &\quad \leq C^2 \sum_{k' \neq k} \int \left| \widehat{p}_{W_{k'},-l} - p_{W_{k'}} \right| d\lambda_{k'} \; = \; o_{\mathbb{P}}(1). \end{aligned}$$

Combining the two parts, for each k,

$$\left\|\widehat{\mathbb{E}}_{-l}\big[\widehat{\mu}_{-l}^2 \mid W_k\big] - \mathbb{E}\Big[\mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2 \mid W_k\Big]\right\| = o_{\mathbb{P}}(1).$$

Similarly, for the second term,

$$\begin{aligned}
&\left|\widehat{\mathbb{E}}_{-l}\big[\widehat{\mu}_{-l}^2\big] - \mathbb{E}\Big[\mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2\Big]\right| \\
&\quad= \left| \begin{aligned} &\int \Big(\widehat{\mu}_{-l}^2 - \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2\Big) \prod_k d\widehat{\mathbb{P}}_{W_k,-l} \\ &\quad + \int \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]^2 \Big(\prod_k d\widehat{\mathbb{P}}_{W_k,-l} - \prod_k d\mathbb{P}_{W_k}\Big) \end{aligned} \right| \\
&\quad\le 2C \left(\int \Big(\widehat{\mu}_{-l} - \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]\Big)^2 \prod_k d\widehat{\mathbb{P}}_{W_k,-l} \right)^{1/2} \\
&\qquad + C^2 \sum_k \int \left|\widehat{p}_{W_k,-l} - p_{W_k}\right| d\lambda_k \\
&\quad\le 2C \left\|\widehat{\mu}_{-l} - \mathbb{E}\big[Y(\boldsymbol{W}) \mid \boldsymbol{W}\big]\right\|_{L_2(\mathbb{P}_{\boldsymbol{W}})} + o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1).
\end{aligned}$$

Since K is fixed, the decomposition term b satisfies $\|b\| \le \sum_{k\in[K]} o_{\mathbb{P}}(1) + K \cdot o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1)$. Thus the decomposition term b and the first-half fitted-score error satisfy $\|b\| = o_{\mathbb{P}}(1)$ and $||(\theta_{\widehat{\mathbb{P}}_{-l}} - \theta)_{\text{first half}}|| \le ||a|| + ||b|| = o_{\mathbb{P}}(1)$.

For the reduced regression,

$$\begin{aligned}
\widehat{\mu}_{-\mathcal{S},-l} &= \int \widehat{\mu}_{-l} \prod_{k\in\mathcal{S}} d\widehat{\mathbb{P}}_{W_k,-l}, \\
\mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}] &= \int \mu \prod_{k\in\mathcal{S}} d\mathbb{P}_{W_k}.
\end{aligned}$$

Conditional Jensen and product telescoping give

$$\begin{aligned}
&\|\widehat{\mu}_{-\mathcal{S},-l} - \mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]\|_{L_2(\mathbb{P}_{\boldsymbol{W}_{-\mathcal{S}}})} \\
&\quad\le \|\widehat{\mu}_{-l} - \mu\|_{L_2(\mathbb{P}_{\boldsymbol{W}})} + C \sum_{k\in\mathcal{S}} t_k \\
&\quad= o_{\mathbb{P}}(n^{-1/4}).
\end{aligned}$$

Thus the reduced learner is bounded and has the required rate without a separate assumption. The same bounds apply to its squares and its products with $\widehat{\mu}_{-l}$.

For the second half,

$$
\begin{aligned}
&(\theta_{\widehat{\mathbb{P}}_{-l}} - \theta)_{\text{second half}} \\
&= \; 2Y(\boldsymbol{W})\widehat{\mu}_{-\mathcal{S},-l} - 2\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l} \\
&\quad + \sum_{j \in -\mathcal{S}} \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-\mathcal{S},-l} \mid W_j] - |-\mathcal{S}| \cdot \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-\mathcal{S},-l}] \\
&+ 2\sum_{i \in \mathcal{S}} \widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l} \mid W_i] - 2|\mathcal{S}| \cdot \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-\mathcal{S},-l}] \\
&- \Big\{ 2Y(\boldsymbol{W})\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] - 2\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] \\
&\quad + \sum_{j \in -\mathcal{S}} \mathbb{E}\big[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2 \mid W_j\big] \\
&- |-\mathcal{S}| \cdot \mathbb{E}\big[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\big] \\
&\quad + 2\sum_{i \in \mathcal{S}} \mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] \mid W_i\right] \\
&- 2|\mathcal{S}| \cdot \mathbb{E}\big[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\big] \Big\} \\
&= \underbrace{\begin{array}{l} 2Y(\boldsymbol{W})\widehat{\mu}_{-\mathcal{S},-l} - 2\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l} \\ \quad - 2Y(\boldsymbol{W})\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] + 2\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] \end{array}}_{:=a} \\
&+ \underbrace{\begin{array}{l} \sum_{j \in -\mathcal{S}} \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-\mathcal{S},-l} \mid W_j] - |-\mathcal{S}| \cdot \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-\mathcal{S},-l}] \\ \quad - \sum_{j \in -\mathcal{S}} \mathbb{E}\big[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2 \mid W_j\big] \end{array}}_{:=b} \\
&\underbrace{+|-\mathcal{S}| \cdot \mathbb{E}\big[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\big]}_{:=b \text{ continued}} \\
&+ \underbrace{\begin{array}{l} 2\sum_{i \in \mathcal{S}} \widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l} \mid W_i] - 2|\mathcal{S}| \cdot \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-\mathcal{S},-l}] \\ \quad - 2\sum_{i \in \mathcal{S}} \mathbb{E}\left[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] \mid W_i\right] \end{array}}_{:=c} \\
&+ \underbrace{2|\mathcal{S}| \cdot \mathbb{E}\big[\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]^2\big]}_{:=c \text{ continued}}.
\end{aligned}
$$

Also, under C1. and C2., we have

$$
\begin{aligned}
||a|| &= \left\| \begin{array}{l} 2Y(\boldsymbol{W})\widehat{\mu}_{-\mathcal{S},-l} - 2\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l} \\ \quad - 2Y(\boldsymbol{W})\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] + 2\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] \end{array} \right\| \\
&= ||2Y(\boldsymbol{W}) \cdot (\widehat{\mu}_{-\mathcal{S},-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]) \\
&\quad - 2\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l} + 2\widehat{\mu}_{-l}\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]
\end{aligned}
$$

$$
\begin{aligned}
&\quad - 2\widehat{\mu}_{-l}\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] + 2\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]|| \\
&= \left\| \begin{matrix} 2(Y(\boldsymbol{W}) - \widehat{\mu}_{-l}) \cdot (\widehat{\mu}_{-\mathcal{S},-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}]) \\ \quad - 2\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}] \cdot (\widehat{\mu}_{-l} - \mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]) \end{matrix} \right\| \\
&\leq 4C\|\widehat{\mu}_{-\mathcal{S},-l} - \mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}]\|_2 + 2C\|\widehat{\mu}_{-l} - \mathbb{E}[Y \mid \boldsymbol{W}]\|_2 \\
&= o_{\mathbb{P}}(1).
\end{aligned}
$$

The remaining decomposition terms satisfy $||b|| = o_{\mathbb{P}}(1)$ and $||c|| = o_{\mathbb{P}}(1)$, where the proofs are similar to that in the first half. Thus the decomposition term c and the second-half fitted-score error satisfy $||c|| = o_{\mathbb{P}}(1)$ and $||(\theta_{\widehat{\mathbb{P}}_{-l}} - \theta)_{\text{second half}}|| \leq ||a|| + ||b|| + ||c|| = o_{\mathbb{P}}(1)$, and the empirical process term is of the order $o_{\mathbb{P}}(1/\sqrt{n})$.

The exact form of the first half of the remainder term $R(\widehat{\mathbb{P}}_{-l}, \mathbb{P}) = \Theta_{\widehat{\mathbb{P}}_{-l}} - \Theta + \mathbb{P}\{\theta_{\widehat{\mathbb{P}}_{-l}}\}$ is

$$
\begin{aligned}
R(\widehat{\mathbb{P}}_{-l}, \mathbb{P})_{\text{first half}} &= \widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}^2] - \int \mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}\right]^2 d\mathbb{P}_{\boldsymbol{W}} \\
&\quad + \int \Big\{ 2Y(\boldsymbol{W})\widehat{\mu}_{-l} - 2\widehat{\mu}_{-l}^2 + \sum_{k\in[K]} \widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}^2 \mid W_k] - K \cdot \widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}^2] \Big\} d\mathbb{P} \\
&= \underbrace{\int \Big\{ 2Y(\boldsymbol{W})\widehat{\mu}_{-l} - \widehat{\mu}_{-l}^2 - \mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}\right]^2 \Big\} d\mathbb{P}}_{:=a} \\
&\quad + \underbrace{\int \Big\{ \sum_{k\in[K]} \widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}^2 \mid W_k] - K \cdot \widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}^2] + \widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}^2] - \widehat{\mu}_{-l}^2 \Big\} d\mathbb{P}_{\boldsymbol{W}}}_{:=b}.
\end{aligned}
$$

Note that, under C1. and C3.,

$$
\begin{aligned}
|a| &= |\int \Big\{ 2Y(\boldsymbol{W})\widehat{\mu}_{-l} - \widehat{\mu}_{-l}^2 - \mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}\right]^2 \Big\} d\mathbb{P}| \\
&= \int (\widehat{\mu}_{-l} - \mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}\right])^2 d\mathbb{P}_{\boldsymbol{W}} \\
&= ||\widehat{\mu}_{-l} - \mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}\right]||^2 \\
&= o_{\mathbb{P}}(1/\sqrt{n}).
\end{aligned}
$$

To bound the first-half remainder term b, define, locally in this proof, the product-law remainder operator for $f \in L_2(\mathbb{P}_{\boldsymbol{W}})$ by

$$
B_f = \sum_{k=1}^{K} \mathbb{E}\{\widehat{\mathbb{E}}_{-l}[f \mid W_k]\} - (K-1)\widehat{\mathbb{E}}_{-l}[f] - \mathbb{E}[f].
$$

In this proof, put $r_k = (p_{W_k} - \widehat{p}_{W_k,-l})/p_{W_k}$ and $e_k = \|r_k\|_{L_2(\mathbb{P}_{W_k})}$. The exact subset expansion of the product measures gives

$$
B_f = -\sum_{\substack{I\subseteq[K] \\ |I|\geq 2}} \mathbb{E}\left[f \prod_{i\in I} r_i \prod_{j\notin I} \frac{\widehat{p}_{W_j,-l}}{p_{W_j}} \right].
$$

Every subset occurs once, including the higher-order terms. Cauchy–Schwarz and product independence imply

$$|B_f| \le \|f\|_{L_2(\mathbb{P}_{\boldsymbol{W}})} \sum_{\substack{I\subseteq[K]\\|I|\ge 2}} \prod_{i\in I} e_i \prod_{j\notin I}(1+e_j),$$
$$e_k \le \underline{c}^{-1}\|p_{W_k} - \widehat{p}_{W_k,-l}\|_{L_2(\mathbb{P}_{W_k})}.$$

Here $\|\widehat{p}_{W_k,-l}/p_{W_k}\|_{L_2(\mathbb{P}_{W_k})} \le 1 + e_k$. For fixed K, every summand has at least two $o_{\mathbb{P}}(n^{-1/4})$ factors. Hence $B_f = o_{\mathbb{P}}(n^{-1/2})$ whenever $\|f\|_{L_2(\mathbb{P}_{\boldsymbol{W}})} = O_{\mathbb{P}}(1)$. The first-half remainder is exactly $-\mathbb{E}[(\widehat{\mu}_{-l} - \mu)^2] + B_{\widehat{\mu}^2_{-l}}$, proving its required rate. For a coordinate subset J and a function f of $\boldsymbol{W}_J$, write B_f^J for the same definition and bound with $[K]$ replaced by J and K by $|J|$.

The empirical-norm statement that motivated Lemma 7 can also be made precise. Set $\Delta_{k,-l} = p_{W_k} - \widehat{p}_{W_k,-l}$ and use the lemma with $h = \Delta_{k,-l}$, restricting C5's class to its nonnegative members. The assumed marginal-density rate gives $\mathbb{P}\Delta^2_{k,-l} \to_{\mathbb{P}} 0$, and C5 supplies membership in a fixed Donsker class. The nonnegative compactness argument in the lemma therefore *derives*, rather than assumes,

$$\|\Delta^2_{k,-l}\|_{L_2(\mathbb{P})} = \|\Delta_{k,-l}\|^2_{L_4(\mathbb{P})} \to_{\mathbb{P}} 0. \tag{22}$$

Consequently

$$\frac{1}{n-n_l}\sum_{i\in\mathcal{I}_{-l}} \Delta_{k,-l}((\boldsymbol{W}_i)_k)^2 = \|\Delta_{k,-l}\|^2_{L_2(\mathbb{P})} + o_{\mathbb{P}}(n^{-1/2}) = o_{\mathbb{P}}(n^{-1/2}). \tag{23}$$

This is an identity for the empirical training measure, not for an arbitrary fitted product law $\widehat{\mathbb{P}}_{\boldsymbol{W},-l}$. The preceding product-law bound does not use this empirical substitution or C5.

For the second half,

$$\begin{aligned}
&R(\widehat{\mathbb{P}}_{-l},\mathbb{P})_{\text{second half}}\\
&= \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-\mathcal{S},-l}] - \int \mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}\right]^2 d\mathbb{P}_{\boldsymbol{W}_{-\mathcal{S}}}\\
&\quad + \int \Big\{ 2Y(\boldsymbol{W})\widehat{\mu}_{-\mathcal{S},-l} - 2\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l}\\
&\qquad + \sum_{j\in-\mathcal{S}} \widehat{\mathbb{E}}_{-l}\left[\widehat{\mu}^2_{-\mathcal{S},-l} \mid W_j\right] - |-\mathcal{S}|\cdot\widehat{\mathbb{E}}_{-l}\left[\widehat{\mu}^2_{-\mathcal{S},-l}\right]\\
&\quad + 2\sum_{i\in\mathcal{S}} \widehat{\mathbb{E}}_{-l}\left[\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l} \mid W_i\right] - 2|\mathcal{S}|\cdot\widehat{\mathbb{E}}_{-l}\left[\widehat{\mu}^2_{-\mathcal{S},-l}\right] \Big\} d\mathbb{P}\\
&= \int \Big\{ 2Y(\boldsymbol{W})\widehat{\mu}_{-\mathcal{S},-l} - \widehat{\mu}^2_{-\mathcal{S},-l} - \mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}\right]^2 \Big\} d\mathbb{P}\\
&\quad + \int \left\{ \begin{array}{l} \displaystyle\sum_{j\in-\mathcal{S}} \widehat{\mathbb{E}}_{-l}\left[\widehat{\mu}^2_{-\mathcal{S},-l} \mid W_j\right] - |-\mathcal{S}|\cdot\widehat{\mathbb{E}}_{-l}\left[\widehat{\mu}^2_{-\mathcal{S},-l}\right]\\ \qquad + \widehat{\mathbb{E}}_{-l}\left[\widehat{\mu}^2_{-\mathcal{S},-l}\right] - \widehat{\mu}^2_{-\mathcal{S},-l} \end{array} \right\} d\mathbb{P}_{\boldsymbol{W}}\\
&\quad + 2\int \left\{ \begin{array}{l} \displaystyle\sum_{k\in[K]} \widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l} \mid W_k] - K\cdot\widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l}]\\ \qquad + \widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l}] - \widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l} \end{array} \right\} d\mathbb{P}_{\boldsymbol{W}}
\end{aligned}$$

$$
\begin{aligned}
&- 2\int \left\{ \begin{array}{l} \sum_{j\in -\mathcal{S}} \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-\mathcal{S},-l} \mid W_j] - |-\mathcal{S}| \cdot \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-\mathcal{S},-l}] \\ \quad + \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-\mathcal{S},-l}] - \widehat{\mu}^2_{-\mathcal{S},-l} \end{array} \right\} d\mathbb{P}_{\boldsymbol{W}} \\
&= \underbrace{\int \left\{ 2Y(\boldsymbol{W})\widehat{\mu}_{-\mathcal{S},-l} - \widehat{\mu}^2_{-\mathcal{S},-l} - \mathbb{E}\left[Y(\boldsymbol{W}) \mid \boldsymbol{W}_{-\mathcal{S}}\right]^2 \right\} d\mathbb{P}}_{:=a} \\
&\quad \underbrace{- \int \left\{ \begin{array}{l} \sum_{j\in -\mathcal{S}} \widehat{\mathbb{E}}_{-l}\left[\widehat{\mu}^2_{-\mathcal{S},-l} \mid W_j\right] - |-\mathcal{S}| \cdot \widehat{\mathbb{E}}_{-l}\left[\widehat{\mu}^2_{-\mathcal{S},-l}\right] \\ \quad + \widehat{\mathbb{E}}_{-l}\left[\widehat{\mu}^2_{-\mathcal{S},-l}\right] - \widehat{\mu}^2_{-\mathcal{S},-l} \end{array} \right\} d\mathbb{P}_{\boldsymbol{W}}}_{:=b} \\
&\quad \underbrace{+ 2\int \left\{ \begin{array}{l} \sum_{k\in [K]} \widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l} \mid W_k] - K \cdot \widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l}] \\ \quad + \widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l}] - \widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l} \end{array} \right\} d\mathbb{P}_{\boldsymbol{W}}}_{:=c}
\end{aligned}
$$

The second equality holds because

$$
\begin{aligned}
&\widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l}] \\
&\quad = \widehat{\mathbb{E}}_{-l}\left[\widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l} \mid \boldsymbol{W}_{-\mathcal{S}}]\right] \\
&\quad = \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-\mathcal{S},-l}]
\end{aligned}
$$

and similarly

$$
\begin{aligned}
&\widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l} \mid W_j] \\
&\quad = \widehat{\mathbb{E}}_{-l}\left[\widehat{\mathbb{E}}_{-l}[\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l} \mid \boldsymbol{W}_{-\mathcal{S}}]|W_j\right] \\
&\quad = \widehat{\mathbb{E}}_{-l}[\widehat{\mu}^2_{-\mathcal{S},-l} \mid W_j],
\end{aligned}
$$

for $j \in -\mathcal{S}$. The displayed identity is equivalently

$$
\begin{aligned}
&R(\widehat{\mathbb{P}}_{-l}, \mathbb{P})_{\text{second half}} \\
&\quad = -\mathbb{E}[(\widehat{\mu}_{-\mathcal{S},-l} - \mathbb{E}[Y \mid \boldsymbol{W}_{-\mathcal{S}}])^2] \\
&\qquad - B^{-\mathcal{S}}_{\widehat{\mu}^2_{-\mathcal{S},-l}} + 2B^{[K]}_{\widehat{\mu}_{-l}\widehat{\mu}_{-\mathcal{S},-l}}.
\end{aligned}
$$

The reduced-regression rate proved above and the subset bound show directly that each term is $o_{\mathbb{P}}(n^{-1/2})$. Thus both numerator components have negligible population remainders; no training empirical-norm substitution is needed.

For the denominator, we denote its efficient influence function by η. For each $k \in [K]$, define the fitted marginal mean and second moment by

$$
\begin{aligned}
\widehat{\mu}_{k,-l}(w_k) &:= \int \widehat{\mu}_{-l}(\boldsymbol{w}) \prod_{j\in[K]\setminus\{k\}} d\widehat{\mathbb{P}}_{W_j,-l}(w_j), \\
\widehat{v}_{k,-l}(w_k) &:= \int \widehat{v}_{-l}(\boldsymbol{w}) \prod_{j\in[K]\setminus\{k\}} d\widehat{\mathbb{P}}_{W_j,-l}(w_j).
\end{aligned}
$$

The exact form of the empirical process term is

$$
\begin{aligned}
&\eta_{\widehat{\mathbb{P}}_{-l}} - \eta \\
=&Y(\boldsymbol{W})^2 - 2Y(\boldsymbol{W})\widehat{\mathbb{E}}_{-l}[Y(\boldsymbol{W})] - \widehat{\upsilon}_{-l} + 2\widehat{\mu}_{-l}\widehat{\mathbb{E}}_{-l}[Y(\boldsymbol{W})] + \sum_{k\in[K]} \widehat{\upsilon}_{k,-l} \\
&- 2\sum_{k\in[K]} \widehat{\mathbb{E}}_{-l}[Y(\boldsymbol{W})]\widehat{\mu}_{k,-l} - K\cdot\widehat{\mathbb{E}}_{-l}[Y(\boldsymbol{W})^2] + 2K\cdot\widehat{\mathbb{E}}_{-l}[Y(\boldsymbol{W})]^2 \\
&- \Big\{Y(\boldsymbol{W})^2 - 2Y(\boldsymbol{W})\mathbb{E}[Y(\boldsymbol{W})] - \mathbb{E}[Y(\boldsymbol{W})^2 \mid \boldsymbol{W}] + 2\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]\mathbb{E}[Y(\boldsymbol{W})]\Big\} \\
&- \Big\{\sum_{k\in[K]} \mathbb{E}[Y(\boldsymbol{W})^2 \mid W_k] - 2\sum_{k\in[K]} \mathbb{E}[Y(\boldsymbol{W})]\mathbb{E}[Y(\boldsymbol{W}) \mid W_k] \\
&\quad - K\cdot\mathbb{E}[Y(\boldsymbol{W})^2] + 2K\cdot\mathbb{E}[Y(\boldsymbol{W})]^2\Big\} \\
=&2Y(\boldsymbol{W})\cdot\Big(\mathbb{E}[Y(\boldsymbol{W})] - \widehat{\mathbb{E}}_{-l}[Y(\boldsymbol{W})]\Big) + \big(\mathbb{E}[Y(\boldsymbol{W})^2 \mid \boldsymbol{W}] - \widehat{\upsilon}_{-l}\big) \\
&+ \Big(2\widehat{\mu}_{-l}\widehat{\mathbb{E}}_{-l}[Y(\boldsymbol{W})] - 2\mathbb{E}[Y(\boldsymbol{W}) \mid \boldsymbol{W}]\mathbb{E}[Y(\boldsymbol{W})]\Big) \\
&+ \sum_{k\in[K]} \big(\widehat{\upsilon}_{k,-l} - \mathbb{E}[Y(\boldsymbol{W})^2 \mid W_k]\big) \\
&\quad + 2\sum_{k\in[K]} \Big(\mathbb{E}[Y(\boldsymbol{W})]\mathbb{E}[Y(\boldsymbol{W}) \mid W_k] - \widehat{\mathbb{E}}_{-l}[Y(\boldsymbol{W})]\widehat{\mu}_{k,-l}\Big) \\
&+ K\cdot\Big(\mathbb{E}[Y(\boldsymbol{W})^2] - \widehat{\mathbb{E}}_{-l}[Y(\boldsymbol{W})^2]\Big) \\
&\quad + 2K\cdot\Big(\widehat{\mathbb{E}}_{-l}[Y(\boldsymbol{W})]^2 - \mathbb{E}[Y(\boldsymbol{W})]^2\Big),
\end{aligned}
$$

To justify consistency when $\widehat{\upsilon}_{-l}$ is not uniformly bounded, C4 gives, for any coordinate subset J and any square-integrable functions f and $\widehat{f}$,

$$
\begin{aligned}
&\left\| \int (\widehat{f} - f) \prod_{j\in J} d\widehat{\mathbb{P}}_{W_j,-l} \right\|_{L_2(\mathbb{P}_{\boldsymbol{W}_{-J}})} \\
&\qquad \le \|\widehat{f} - f\|_{L_2(\mathbb{P}_{\boldsymbol{W}})} \prod_{j\in J}(1 + e_j).
\end{aligned}
$$

For bounded f, product telescoping also gives

$$
\begin{aligned}
&\left\| \int f \left(\prod_{j\in J} d\widehat{\mathbb{P}}_{W_j,-l} - \prod_{j\in J} d\mathbb{P}_{W_j} \right) \right\|_2 \\
&\qquad \le \|f\|_\infty \sum_{j\in J} t_j.
\end{aligned}
$$

Apply these inequalities to $f = \upsilon$, which is bounded by C^2, and $\widehat{f} = \widehat{\upsilon}_{-l}$. They prove consistency of every marginal second moment in the variance score. The analogous bounds for the means, together with $\widehat{\mathbb{E}}_{-l}[Y] \to_{\mathbb{P}} \mathbb{E}[Y]$, prove $\|\eta_{\widehat{\mathbb{P}}_{-l}} - \eta\|_2 = o_{\mathbb{P}}(1)$.

We then move on to the remainder term:

$$
\begin{aligned}
&\mathsf{Var}(Y)_{\widehat{\mathbb{P}}_{-l}} - \mathsf{Var}(Y) + \mathbb{P}\{\eta_{\widehat{\mathbb{P}}_{-l}}\} \\
=&\widehat{\mathbb{E}}_{-l}[Y^2] - \widehat{\mathbb{E}}_{-l}[Y]^2 - \mathbb{E}[Y^2] + \mathbb{E}[Y]^2 \\
&+ \int \Big\{ Y^2 - 2Y \cdot \widehat{\mathbb{E}}_{-l}[Y] - \widehat{\upsilon}_{-l} + 2\widehat{\mu}_{-l} \cdot \widehat{\mathbb{E}}_{-l}[Y] + \sum_{k\in[K]} \widehat{\upsilon}_{k,-l} \\
&- 2 \sum_{k\in[K]} \widehat{\mu}_{k,-l} \cdot \widehat{\mathbb{E}}_{-l}[Y] - K \cdot \widehat{\mathbb{E}}_{-l}[Y^2] + 2K \cdot \widehat{\mathbb{E}}_{-l}[Y]^2 \Big\} d\mathbb{P} \\
=& \underbrace{\int \Big\{ \mathbb{E}[Y]^2 - 2Y \cdot \widehat{\mathbb{E}}_{-l}[Y] + \widehat{\mathbb{E}}_{-l}[Y]^2 \Big\} d\mathbb{P}}_{:=a} \\
&\quad + \underbrace{\int \Big\{ \widehat{\mathbb{E}}_{-l}[Y^2] - \widehat{\upsilon}_{-l} + \sum_{k\in[K]} \widehat{\upsilon}_{k,-l} - K \cdot \widehat{\mathbb{E}}_{-l}[Y^2] \Big\} d\mathbb{P}}_{:=b} \\
&+ \underbrace{\int \left\{ \begin{array}{l} 2\widehat{\mu}_{-l} \cdot \widehat{\mathbb{E}}_{-l}[Y] - 2\widehat{\mathbb{E}}_{-l}[Y]^2 \\ \quad + 2K \cdot \widehat{\mathbb{E}}_{-l}[Y]^2 - 2 \sum_{k\in[K]} \widehat{\mu}_{k,-l} \cdot \widehat{\mathbb{E}}_{-l}[Y] \end{array} \right\} d\mathbb{P}}_{:=c}.
\end{aligned}
$$

For the coherent fitted mean in Table 4,

$$
\widehat{\mathbb{E}}_{-l}[Y] = \int \widehat{\mu}_{-l} \, d\widehat{\mathbb{P}}_{\boldsymbol{W},-l},
$$

rather than an unrelated empirical mean,

$$
\begin{aligned}
|\widehat{\mathbb{E}}_{-l}[Y] - \mathbb{E}[Y]| &\le \|\widehat{\mu}_{-l} - \mu\|_{L_2(\mathbb{P}_{\boldsymbol{W}})} + C \sum_k t_k \\
&= o_{\mathbb{P}}(n^{-1/4}).
\end{aligned}
$$

The preceding decomposition is exactly

$$
\begin{aligned}
&R(\widehat{\mathbb{P}}_{-l}, \mathbb{P})_{\text{denominator}} \\
&\quad = (\widehat{\mathbb{E}}_{-l}[Y] - \mathbb{E}[Y])^2 + B_{\widehat{\upsilon}_{-l}} \\
&\qquad - 2\widehat{\mathbb{E}}_{-l}[Y] B_{\widehat{\mu}_{-l}}.
\end{aligned}
$$

Since $\|\widehat{\upsilon}_{-l} - \upsilon\|_2 = o_{\mathbb{P}}(n^{-1/4})$ and $|\upsilon| \le C^2$, $\|\widehat{\upsilon}_{-l}\|_2 = O_{\mathbb{P}}(1)$. The weighted subset bound therefore applies to $\widehat{\upsilon}_{-l}$ without a sup-norm assumption, and gives $R(\widehat{\mathbb{P}}_{-l}, \mathbb{P})_{\text{denominator}} = o_{\mathbb{P}}(n^{-1/2})$. The actual fitted second moment also satisfies

$$
\begin{aligned}
|\widehat{\mathbb{E}}_{-l}[Y^2] - \mathbb{E}[Y^2]| &\le \|\widehat{\upsilon}_{-l} - \upsilon\|_2 \prod_k (1 + e_k) + C^2 \sum_k t_k \\
&= o_{\mathbb{P}}(n^{-1/4}).
\end{aligned}
$$

It follows that $\mathsf{Var}(Y)_{\widehat{\mathbb{P}}_{-l}} - \mathsf{Var}(Y) = o_{\mathbb{P}}(n^{-1/4})$. Bounded regression learners, their quarter rates, and the same product-law bounds give the quarter rate of each fitted regression-square numerator. In particular, the fitted denominator converges to $\mathsf{Var}(Y) > 0$.

It remains to connect the component expansions to the direct quotient correction in Algorithm 1. On fold l, use the uncorrected component plug-ins $\Theta_{\widehat{\mathbb{P}}_{-l}}$ and $\mathsf{Var}(Y)_{\widehat{\mathbb{P}}_{-l}}$. Locally in this proof, let $u_l = \mathbb{P}_n^l\{\theta_{\widehat{\mathbb{P}}_{-l}}\}$ and $v_l = \mathbb{P}_n^l\{\eta_{\widehat{\mathbb{P}}_{-l}}\}$ denote their empirical score corrections. The preceding population remainders and centered empirical bounds imply

$$\begin{aligned}\Theta_{\widehat{\mathbb{P}}_{-l}} + u_l - \Theta &= (\mathbb{P}_n^l - \mathbb{P})\theta + o_{\mathbb{P}}(n^{-1/2}),\\ \mathsf{Var}(Y)_{\widehat{\mathbb{P}}_{-l}} + v_l - \mathsf{Var}(Y) &= (\mathbb{P}_n^l - \mathbb{P})\eta + o_{\mathbb{P}}(n^{-1/2}).\end{aligned}$$

Together with the plug-in quarter rates, these identities give empirical score corrections $u_l, v_l = o_{\mathbb{P}}(n^{-1/4})$. Whenever the denominators are nonzero, the exact identity

$$\begin{aligned}&\frac{\Theta_{\widehat{\mathbb{P}}_{-l}} + u_l}{\mathsf{Var}(Y)_{\widehat{\mathbb{P}}_{-l}} + v_l}\\ &\quad - \left(\frac{\Theta_{\widehat{\mathbb{P}}_{-l}}}{\mathsf{Var}(Y)_{\widehat{\mathbb{P}}_{-l}}} + \frac{u_l}{\mathsf{Var}(Y)_{\widehat{\mathbb{P}}_{-l}}} - \frac{\Theta_{\widehat{\mathbb{P}}_{-l}} v_l}{\{\mathsf{Var}(Y)_{\widehat{\mathbb{P}}_{-l}}\}^2}\right)\\ &\quad = \frac{v_l(\Theta_{\widehat{\mathbb{P}}_{-l}} v_l - \mathsf{Var}(Y)_{\widehat{\mathbb{P}}_{-l}} u_l)}{\{\mathsf{Var}(Y)_{\widehat{\mathbb{P}}_{-l}}\}^2(\mathsf{Var}(Y)_{\widehat{\mathbb{P}}_{-l}} + v_l)}\\ &\quad = o_{\mathbb{P}}(n^{-1/2})\end{aligned}$$

shows that the direct correction has the same first-order expansion as the ratio of corrected components. The delta method gives the foldwise true influence function $\varphi = \theta/\mathsf{Var}(Y) - \Theta\eta/\{\mathsf{Var}(Y)\}^2$. Using the actual weights in the algorithm,

$$\widehat{\xi} - \xi = \sum_{l=1}^{L} \frac{n_l}{n}(\mathbb{P}_n^l - \mathbb{P})\varphi + o_{\mathbb{P}}(n^{-1/2}) = \frac{1}{n}\sum_{i=1}^{n} \varphi(Y_i, \boldsymbol{W}_i; \mathbb{P}) + o_{\mathbb{P}}(n^{-1/2}).$$

No independence between the overlapping training folds is used: there are finitely many negligible fold remainders, and the leading sum is over the original i.i.d. observations. The i.i.d. CLT proves the stated normal limit.

Finally, this argument also justifies the actual variance estimator in Algorithm 1. The preceding component-score consistency and convergence of the fitted quotient coefficients imply $\|\varphi_{\widehat{\mathbb{P}}_{-l}} - \varphi\|_{L_2(\mathbb{P})} \to_{\mathbb{P}} 0$. Put $Z_i = \xi + \varphi(Y_i, \boldsymbol{W}_i; \mathbb{P})$. Conditional Markov's inequality on each held-out fold, together with the plug-in consistency, gives

$$\frac{1}{n}\sum_{i=1}^{n}(\widehat{\xi}_i - Z_i)^2 \to_{\mathbb{P}} 0.$$

For this step one may bound the conditional probability by the minimum of one and the conditional squared error divided by a fixed threshold; its expectation tends to zero. Thus no unscaled expected-risk assumption is required. The law of large numbers for Z_i and Z_i^2, followed by Cauchy–Schwarz,

shows that the empirical first and second moments of $\widehat{\xi}_i$ converge to those of Z_i. Hence

$$\widehat{\sigma}^2 = \frac{1}{n}\sum_i (\widehat{\xi}_i - \widehat{\xi})^2 \to_{\mathbb{P}} \mathsf{Var}\{\varphi(Y, \boldsymbol{W}; \mathbb{P})\}.$$

When this limit is positive, Slutsky's theorem yields studentization and the coverage asserted in Corollary 2. The same variance-consistency argument applies to the baseline influence function in (9).

■

D.5. Proof of Proposition 5

Proof In this proof, let $D = Y(\boldsymbol{W}) - Y(\boldsymbol{W}'_{\mathcal{S}}, \boldsymbol{W}_{-\mathcal{S}})$. The two outcomes have the same distribution by Equation (1) and Assumption 1, so $\mathbb{E}[D] = 0$. Since $\mathsf{Var}(Y) > 0$,

$$\xi(\vee_{k\in\mathcal{S}} W_k) = 0 \quad \Longleftrightarrow \quad \mathsf{Var}(D) = 0 \quad \Longleftrightarrow \quad D = 0 \text{ almost surely.} \tag{24}$$

For discrete assignments, each supported pair has positive probability, giving the stated equality almost surely over potential outcomes at that pair. For general assignments the conclusion is almost sure, not pointwise outside the assignment law. ■

D.6. Proof of Proposition 6

Proof Under the sharp null, the full dataset $(\boldsymbol{W}_{\mathcal{S}}, \boldsymbol{W}_{-\mathcal{S}}, Y)$ is invariant in law under permutations of the rows of $\boldsymbol{W}_{\mathcal{S}}$: the treatment block is independent of the retained treatments and null outcomes, and the units are i.i.d. Conditional on its permutation orbit and on $(\boldsymbol{W}_{-\mathcal{S}}, Y)$, the observed treatment array is uniform on that orbit. Independent uniform draws from the full permutation group therefore make the observed and B resampled statistics jointly exchangeable, using the same analysis rule for all datasets. The plus-one rank p-value, with weak inequalities for ties, consequently satisfies $\mathbb{P}(P \le \alpha) \le \alpha$. ■

D.7. Proof of Proposition 7

Proof In this proof, let R_n be first-stage rejection and I_n the second-stage interval in (12). At total explainability $\xi(\vee_{k\in\mathcal{S}} W_k) = 0$, noncoverage is contained in R_n, so its probability is at most α by Proposition 6. At each fixed positive total explainability $\xi(\vee_{k\in\mathcal{S}} W_k) > 0$, noncoverage of the sequential procedure is contained in $R_n^c \cup \{\xi(\vee_{k\in\mathcal{S}} W_k) \notin I_n\}$. The first probability tends to zero by power consistency and the second has limit superior at most α by Corollary 2. No independence between the stages is required. With the fallback $[0, \infty)$ on nonrejection, noncoverage at a positive alternative is contained in $\{\xi(\vee_{k\in\mathcal{S}} W_k) \notin I_n\}$, so the power condition is unnecessary. At the null, noncoverage remains contained in R_n and is bounded by α, not necessarily zero. ■

Appendix E. Additional tables

Expression	**Meaning**
$[K]$	Index set $\{1, \ldots, K\}$ for the K factors
Y	Outcome
$\boldsymbol{W} = (W_1, \ldots, W_K)$	Vector of factors
$\boldsymbol{W}_{\mathcal{S}}$	Subvector of $\boldsymbol{W}$ indexed by $\mathcal{S} \subseteq [K]$
W_k	k-th component of $\boldsymbol{W}$
$\mathbb{P}$/$\widehat{\mathbb{P}}$	True/estimated joint distribution of $(Y, \boldsymbol{W})$
$\mathbb{P}_{\boldsymbol{W}}$/$\widehat{\mathbb{P}}_{\boldsymbol{W}}$	True/estimated marginal distribution of $\boldsymbol{W}$
$\mathbb{P}_{W_k}$/$\widehat{\mathbb{P}}_{W_k}$	True/estimated marginal distribution of W_k
$\mathbb{P}_{Y\|\boldsymbol{W}}$/$\widehat{\mathbb{P}}_{Y\|\boldsymbol{W}}$	True/estimated conditional distribution of Y given $\boldsymbol{W}$
$\mathbb{E}[\cdot]$/$\widehat{\mathbb{E}}[\cdot]$	Expectation under $\mathbb{P}$/$\widehat{\mathbb{P}}$
$\mu(\boldsymbol{w})$, $\upsilon(\boldsymbol{w})$	Conditional mean and second moment given $\boldsymbol{W} = \boldsymbol{w}$
$\mathcal{P}$	Probability model (set of data-generating distributions)
$\mathcal{P}_{\text{np}}$	Non-parametric probability model (no structural restrictions)
$\mathcal{P}_{\text{ind}}$	Probability model with independent components of $\boldsymbol{W}$
$\dot{\mathcal{P}}_{\mathbb{P}}$	Tangent space at $\mathbb{P} \in \mathcal{P}$
ξ	Causal estimand of interest
$\xi(\vee_{k\in\mathcal{S}} W_k)$, $\mathcal{S} \subseteq [K]$	Causal explainability attributable to the union of factors in $\mathcal{S}$
$\xi(W_k \wedge W_{k'})$	Causal explainability attributable to the interaction of W_k and $W_{k'}$
$\varphi^{\xi}_{\text{IF}}$, $\varphi^{\xi}_{\text{EIF}}$, φ^{ξ}	(Efficient) influence functions for ξ

Table 3: Table of notations. When unambiguous, superscripts or subscripts may be omitted.

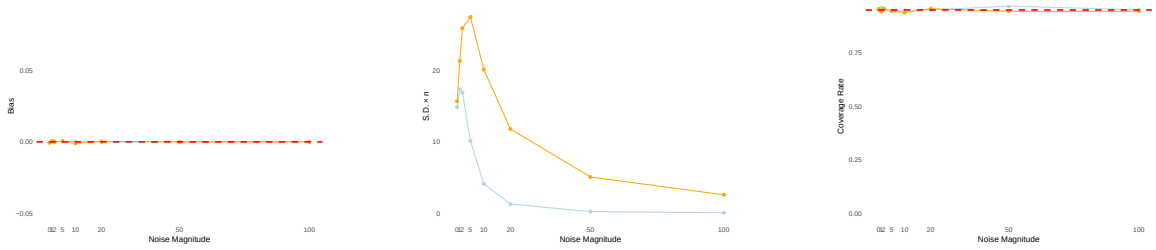
Expression	Formula
$\widehat{\mathbb{E}}[Y \mid \boldsymbol{W} = \boldsymbol{w}]$	$\widehat{\mu}(\boldsymbol{w})$
$\widehat{\mathbb{E}}[Y \mid \boldsymbol{W}_{-\mathcal{S}} = \boldsymbol{w}_{-\mathcal{S}}]$	$\int \widehat{\mu}(\boldsymbol{w}) d\widehat{\mathbb{P}}_{\boldsymbol{W}_{\mathcal{S}}}(\boldsymbol{w}_{\mathcal{S}})$
$\widehat{\mathbb{E}}[\widehat{\mathbb{E}}^2[Y \mid \boldsymbol{W}_{-\mathcal{S}}]]$	$\int \left(\int \widehat{\mu}(\boldsymbol{w}) d\widehat{\mathbb{P}}_{\boldsymbol{W}_{\mathcal{S}}}(\boldsymbol{w}_{\mathcal{S}})\right)^2 d\widehat{\mathbb{P}}_{\boldsymbol{W}_{-\mathcal{S}}}(\boldsymbol{w}_{-\mathcal{S}})$
$\widehat{\mathbb{E}}\left[\widehat{\mathbb{E}}[Y \mid \boldsymbol{W}] \cdot \widehat{\mathbb{E}}[Y \mid \boldsymbol{W}_{-\mathcal{S}}] \mid W_i = w_i\right]$, $i \in \mathcal{S}$	$\int \widehat{\mu}(\boldsymbol{w}) \cdot \left(\int \widehat{\mu}(\boldsymbol{w}) d\widehat{\mathbb{P}}_{\boldsymbol{W}_{\mathcal{S}}}(\boldsymbol{w}_{\mathcal{S}})\right) d\widehat{\mathbb{P}}_{\boldsymbol{W}_{-i}}(\boldsymbol{w}_{-i})$
$\widehat{\mathbb{E}}\left[\widehat{\mathbb{E}}^2[Y \mid \boldsymbol{W}_{-\mathcal{S}}] \mid W_j = w_j\right]$, $j \notin \mathcal{S}$	$\int \left(\int \widehat{\mu}(\boldsymbol{w}) d\widehat{\mathbb{P}}_{\boldsymbol{W}_{\mathcal{S}}}(\boldsymbol{w}_{\mathcal{S}})\right)^2 d\widehat{\mathbb{P}}_{\boldsymbol{W}_{-j}}(\boldsymbol{w}_{-j})$
$\widehat{\mathbb{E}}_{-\ell}[Y]$, φ_{EIF}	$\int \widehat{\mu}(\boldsymbol{w})\, d\widehat{\mathbb{P}}_{\boldsymbol{W}}(\boldsymbol{w})$
$\widehat{\mathbb{E}}_{-\ell}[Y^2]$, φ_{EIF}	$\int \widehat{\upsilon}(\boldsymbol{w})\, d\widehat{\mathbb{P}}_{\boldsymbol{W}}(\boldsymbol{w})$
$\mathsf{Var}(Y)_{\widehat{\mathbb{P}}_{-\ell}}$, φ_{IF}	$(n - n_\ell)^{-1} \sum_{i\in\mathcal{I}_{-\ell}} (Y_i - \widehat{\mathbb{E}}_{-\ell}[Y])^2$
$\mathsf{Var}(Y)_{\widehat{\mathbb{P}}_{-\ell}}$, φ_{EIF}	$\int \widehat{\upsilon}(\boldsymbol{w}) d\widehat{\mathbb{P}}_{\boldsymbol{W}}(\boldsymbol{w}) - \left(\int \widehat{\mu}(\boldsymbol{w}) d\widehat{\mathbb{P}}_{\boldsymbol{W}}(\boldsymbol{w})\right)^2$

Table 4: Formulas for the one-step correction estimator. The marginal-integration formulas use the fitted product law. Fitted regressions and treatment marginals are trained without fold ℓ; their fold subscripts are omitted inside the integrals.

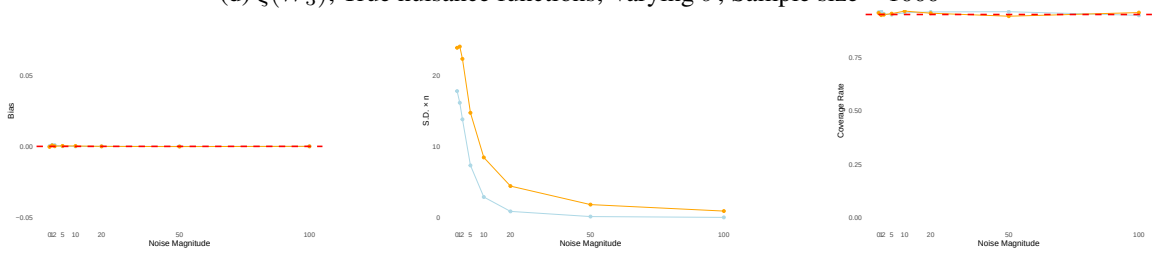
Appendix F. Additional empirical studies

F.1. Additional simulations with true nuisances

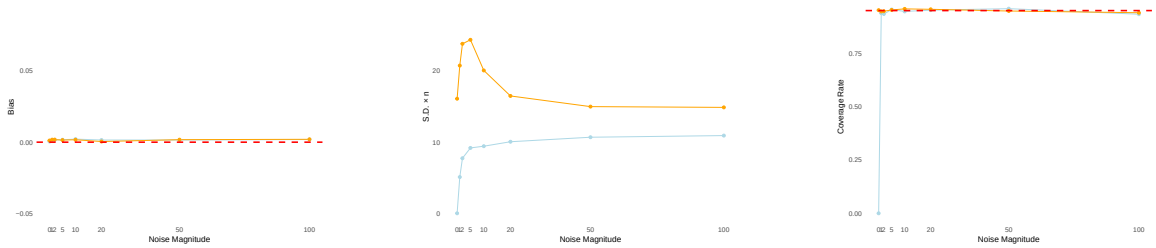
We implement the two methods using the true nuisance functions. In Figure 4, we vary the interaction coefficient σ.



(d) $\xi(W_3)$; True nuisance functions; Varying σ; Sample size = 1000



(e) $\xi(W_1 \vee W_3)$; True nuisance functions; Varying σ; Sample size = 1000



(f) $\xi(W_1 \wedge W_3)$; True nuisance functions; Varying σ; Sample size = 1000

Figure 4: Comparison of bias (left panel), estimated standard deviation times sample size (mid panel), and coverage rate (right panel; significance level $\alpha = 0.05$) with varying σ. φ_{EIF}-based method in blue, φ_{IF}-based method in gold. True nuisance functions are used. Results aggregated over 1000 trials.

F.2. Additional simulations with estimated nuisances

We compared φ_{EIF}-based estimator and the φ_{IF}-based estimator for total explainability in Section 5, and here we focus on the more challenging problem of interaction explainability. Owing to the increased complexity of nuisance function estimation inherent in interaction effects, both methods exhibit coverage below the target level. Recall that φ_{EIF}-based estimator requires estimating more complicated nuisance components to exploit independence; in this setting, the theoretical gain in

asymptotic variance is offset by the increased variability from nuisance estimation. As a result, the φ_{EIF}-based estimator appears less favorable in terms of both MSE and coverage.

Table 5: Performance comparison of φ_{EIF} and φ_{IF}-based estimators with estimated nuisance function regarding interaction expliainability. Aggregated over 100 trials.

Method	Bias	SD $\times n$	Coverage ($\alpha = 0.05$)
φ_{EIF}	-0.11	2.87	0.79
φ_{IF}	0.07	2.58	0.83

Appendix G. Details of the immigration experiment

The attributes include each immigrant's gender, education level, employment plans, job experience, profession, language skills, country of origin, reasons for applying, and prior trips to the United States. Respondents' characteristics such as age, education, ethnicity, gender, and ethnocentrism are also available in the survey.

One thing should be noted is that there are two restrictions imposed on the possible combinations of immigrant attributes. First, those immigrant profiles who were fleeing persecution were restricted to come from countries where such an application was plausible (e.g., Iraq, Sudan). Second, those immigrants occupying high-skill occupations (e.g., research scientists, doctors) were constrained to have at least two years of college education. In this case, the distribution of immigrants' occupations is dependent on their education, but conditionally independent of the other attributes and the other profiles in the same choice task given their education (Hainmueller et al., 2015). However, as this is an artificial experiment, it is very unclear whether the attributes are *causally* dependent or not. The reason is that the restrictions are put in the design stage, so attributes are simultaneously randomized at the moment the profile is generated. To avoid the problem of unknown directions in a directed acyclic graph, we choose to collapse the two pairs of dependent variables into two variables —— *Origin & Application reason* and *Education & Profession*. Another thing is that although it is a randomized experiment, the authors did not provide any information on the randomization mechanism. So we do the analysis assuming that the propensity scores are unknown.

Respondent-level inference. In the respondent-level analysis of Section 6, let $O_{ir} = (Y_{ir}, \boldsymbol{W}_{ir})$ denote row r of respondent i, and let $O_i = \{O_{ir}\}_{r=1}^{m_i}$ denote all observations from respondent i. We assume $\{O_i\}_{i=1}^{n}$ are i.i.d. across i, and allow arbitrary dependence within i. The cluster-robust confidence intervals in Section 6 use an asymptotic approximation, requiring a cluster-level asymptotic linearization and finite positive variance of the limiting respondent score. Here each respondent contributes ten profile rows; with variable or response-dependent cluster sizes, row normalization instead requires the corresponding row-weighted target and centering and moment conditions. Any randomization test must preserve the actual assignment mechanism. Under clustering, we resample/permutate $\boldsymbol{W}_{\mathcal{S}}$ within the respondent (and within blocks/strata if present), preserving task structure and any design constraints (e.g., paired-profile choice sets, restricted attribute combinations). We then recompute the test statistic on each resampled dataset using the same respondent-level cross-fitting and cluster-robust studentization. Finite-sample randomization validity requires that the resampling scheme match the true constrained assignment law and that the observed and all re-

sampled statistics be jointly exchangeable. Because the full assignment mechanism is unavailable, design-exact validity of the reported empirical permutation results is conditional on this requirement.

Table 6: Conditional Randomization Test (Ham et al., 2024)

Interaction	p **value**
Gender & Job Experience	0.9108911
Job Plan & Gender	0.6831683
Job Plan & Language	0.0990099
Job Plan & Job Experience	0.00990099**
Job Plan & Prior trips to U.S.	0.5049505

Table 7: Average Component Interaction Effects (ACIE) (Hainmueller et al., 2015)

Attribute	**Estimate**	**Std. Err**	**z value**	**Pr**($>\|z\|$)
Gender:Job Experience				
male:1-2 years	-0.0017453	0.023181	-0.07529	0.93998
male:3-5 years	-0.0215082	0.024037	-0.89481	0.37089
male:5+ years	-0.0088778	0.024508	-0.36224	0.71717
Gender:Job Plan				
male:contract with employer	0.0034950	0.023374	0.14953	0.88114
male:interviews with employer	-0.0185813	0.024573	-0.75616	0.44955
male:no plans to look for work	0.0042381	0.022811	0.18579	0.85261
Job Plan:Language				
contract with employer:broken English	0.0477691	0.031921	1.49649	0.134527
interviews with employer:broken English	0.0652677	0.033518	1.94725	0.051505
no plans to look for work:broken English	0.0117892	0.032696	0.36057	0.718423
contract with employer:tried English but unable	0.0100675	0.032462	0.31013	0.756463
interviews with employer:tried English but unable	0.0274033	0.032797	0.83555	0.403405
no plans to look for work:tried English but unable	0.0069235	0.031912	0.21696	0.828243
contract with employer:used interpreter	0.0150713	0.032272	0.46700	0.640497
interviews with employer:used interpreter	0.0499209	0.032747	1.52444	0.127399
no plans to look for work:used interpreter	0.0402530	0.031193	1.29046	0.196892

Number of Obs. = 13960, Number of Respondents = 1396.

Significance codes: 0 *** 0.001 ** 0.01 * 0.05

Table 8: Average Component Interaction Effects (ACIE)-continued

Attribute	**Estimate**	**Std. Err**	**z value**	**Pr($>\|z\|$)**
Job Experience:Job Plan				
1-2 years:contract with employer	0.03528079	0.032295	1.092449	0.274636
3-5 years:contract with employer	0.00069369	0.033193	0.020899	0.983327
5+ years:contract with employer	-0.02503323	0.033785	-0.740960	0.458718
1-2 years:interviews with employer	0.01152705	0.033719	0.341853	0.732462
3-5 years:interviews with employer	-0.02887657	0.034100	-0.846821	0.397095
5+ years:interviews with employer	-0.03515407	0.034809	-1.009902	0.312542
1-2 years:no plans to look for work	-0.00947529	0.031805	-0.297922	0.765762
3-5 years:no plans to look for work	-0.07701162	0.032256	-2.387519	0.016963*
5+ years:no plans to look for work	-0.06210952	0.032943	-1.885391	0.059377
Job Plan:Prior Entry				
contract with employer:once as tourist	-0.0306128	0.036545	-0.83767	0.402218
interviews with employer:once as tourist	0.0639714	0.037340	1.71320	0.086676
no plans to look for work:once as tourist	0.0469911	0.037612	1.24936	0.211533
contract with employer:many times as tourist	-0.0034264	0.036436	-0.09404	0.925077
interviews with employer:many times as tourist	0.0075229	0.038365	0.19609	0.844544
no plans to look for work:many times as tourist	0.0355192	0.037469	0.94797	0.343144
contract with employer:six months with family	0.0191688	0.036835	0.52039	0.602792
interviews with employer:six months with family	0.0508093	0.037914	1.34012	0.180206
no plans to look for work:six months with family	0.0392413	0.036617	1.07166	0.283871
contract with employer:once w/o authorization	0.0091775	0.036384	0.25224	0.800859
interviews with employer:once w/o authorization	0.0302793	0.037539	0.80661	0.419892
no plans to look for work:once w/o authorization	0.0620386	0.035214	1.76175	0.078112

Number of Obs. = 13960, Number of Respondents = 1396.

Significance codes: 0 *** 0.001 ** 0.01 * 0.05

Table 9: Average Component Interaction Effects (ACIE)-continued

Attribute	**Estimate**	**Std. Err**	**z value**	**Pr($>\|z\|$)**
Job:Job Plan				
waiter:contract with employer	-0.0327390	0.047473	-0.689639	0.490421
child care provider:contract with employer	-0.0900152	0.050735	-1.774211	0.076028
gardener:contract with employer	0.0311797	0.048505	0.642814	0.520345
financial analyst:contract with employer	-0.0484468	0.085065	-0.569525	0.569000
construction worker:contract with employer	-0.0493627	0.049477	-0.997686	0.318432
teacher:contract with employer	-0.0363747	0.050976	-0.713561	0.475499
computer programmer:contract with employer	-0.1469473	0.085449	-1.719704	0.085486
nurse:contract with employer	-0.0603685	0.047341	-1.275183	0.202245
research scientist:contract with employer	-0.1356852	0.080691	-1.681531	0.092660
doctor:contract with employer	-0.1538243	0.077780	-1.977672	0.047966*
waiter:interviews with employer	-0.0450223	0.049338	-0.912537	0.361486
child care provider:interviews with employer	-0.0383598	0.049890	-0.768895	0.441956
gardener:interviews with employer	-0.0413542	0.050545	-0.818161	0.413265
financial analyst:interviews with employer	-0.0863427	0.092088	-0.937614	0.348443
construction worker:interviews with employer	-0.0211739	0.051522	-0.410968	0.681096
teacher:interviews with employer	0.0301964	0.050484	0.598142	0.549746
computer programmer:interviews with employer	-0.1196731	0.088597	-1.350757	0.176773
nurse:interviews with employer	-0.0116562	0.049813	-0.234000	0.814985
research scientist:interviews with employer	-0.0679747	0.084104	-0.808220	0.418964
doctor:interviews with employer	-0.1211354	0.082133	-1.474867	0.140248
waiter:no plans to look for work	0.0028488	0.047582	0.059871	0.952258
child care provider:no plans to look for work	0.0035945	0.048850	0.073583	0.941343
gardener:no plans to look for work	0.0303911	0.048098	0.631859	0.527479
financial analyst:no plans to look for work	0.0443680	0.086945	0.510300	0.609842
construction worker:no plans to look for work	-0.0190767	0.048589	-0.392611	0.694607
teacher:no plans to look for work	0.0125936	0.049917	0.252292	0.800815
computer programmer:no plans to look for work	0.0012924	0.083344	0.015506	0.987628
nurse:no plans to look for work	-0.0131314	0.047923	-0.274010	0.784077
research scientist:no plans to look for work	0.0331237	0.085202	0.388766	0.697449
doctor:no plans to look for work	-0.0040783	0.079754	-0.051136	0.959217

Number of Obs. = 13960, Number of Respondents = 1396.

Significance codes: 0 *** 0.001 ** 0.01 * 0.05